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(54) **PERFORMANCE MONITORING AND CONTROL SYSTEM FOR CONNECTED BUILDING EQUIPMENT WITH STABILITY INDEX**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,873,649 A * 10/1989 Grald F24F 11/46
62/226
5,083,438 A 1/1992 McMullin
(Continued)

OTHER PUBLICATIONS

Al-Sharafi et al., "Overall performance index for hybrid power plants," Energy Conversion and Management, 2015, vol. 100 (pp. 103-116).

(Continued)

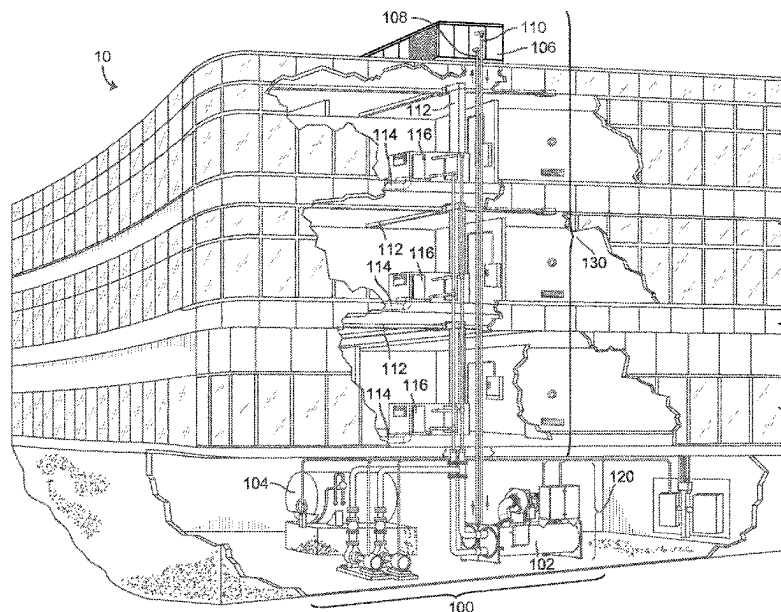
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(57) **ABSTRACT**

A method includes obtaining an actual value of a condition affected by operating the building equipment, operating the building equipment using a setpoint for the condition, calculating a stability index based on a timeseries of errors between the actual value and the setpoint, comparing the stability index to a criterion, in response to the stability index satisfying the criterion, executing an action relating to the building equipment, and in response to the stability index not satisfying the criterion, not executing the action.

20 Claims, 13 Drawing Sheets



Related U.S. Application Data

a continuation-in-part of application No. 17/028,704, filed on Sep. 22, 2020, now Pat. No. 11,519,620, said application No. 17/400,950 is a continuation of application No. 16/245,122, filed on Jan. 10, 2019, now Pat. No. 11,092,954.

(60) Provisional application No. 63/298,720, filed on Jan. 12, 2022.

(52) U.S. Cl.

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(58) Field of Classification Search

CPC G05B 2219/2642; G05B 23/024; G05B 23/0286; G05B 2219/2614; G05B 15/02
See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

10,257,052	B1	4/2019	Nborg et al.
10,290,025	B1 *	5/2019	Howes G06Q 30/0261
10,378,794	B2	8/2019	Togano et al.
11,092,954	B2	8/2021	Vitullo et al.
11,493,913	B2	11/2022	Shukla et al.
2006/0144057	A1	7/2006	You et al.
2009/0140060	A1 *	6/2009	Stoner G05D 23/1934

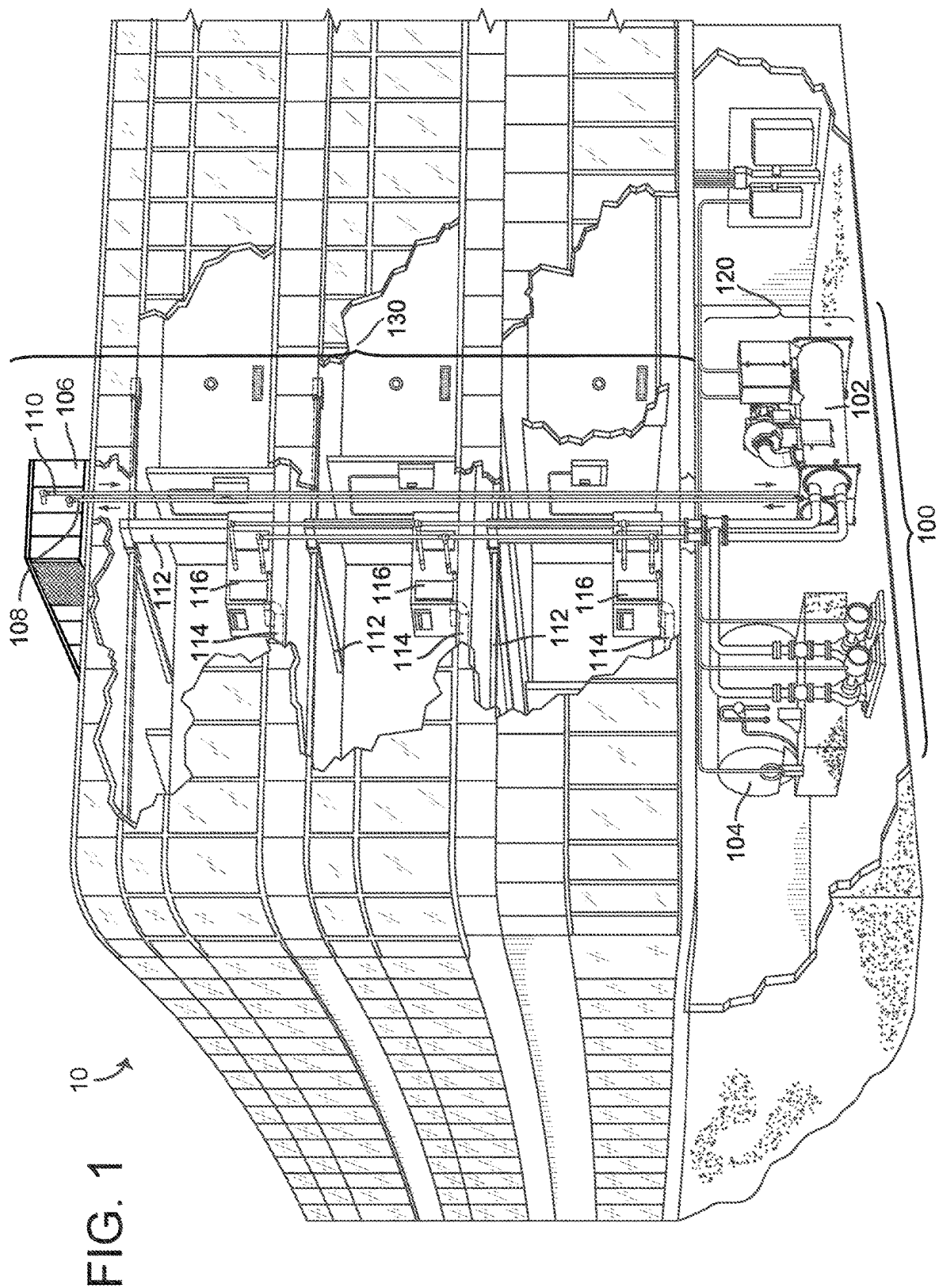
236/51

2009/0210081	A1	8/2009	Sustaeta et al.
2012/0259583	A1 *	10/2012	Noboa H02J 13/00002
			702/179
2013/0098084	A1	4/2013	Matsuo et al.
2015/0167999	A1	6/2015	Seem et al.
2015/0178865	A1	6/2015	Anderson et al.
2015/0316907	A1	11/2015	Elbsat et al.
2016/0057029	A1	2/2016	Iida et al.
2016/0061693	A1	3/2016	Salsbury et al.
2016/0334122	A1	11/2016	Shiel
2017/0276571	A1	9/2017	Vitullo et al.
2017/0293293	A1	10/2017	Brownie et al.
2017/0309094	A1	10/2017	Farahat et al.
2017/0366414	A1	12/2017	Hamilton et al.
2018/0039292	A1 *	2/2018	van Berkel F24H 15/148
2018/0046151	A1	2/2018	Ahmed
2018/0087790	A1	3/2018	Perez
2018/0373234	A1	12/2018	Khalate et al.
2019/0086108	A1 *	3/2019	Okita F24F 11/46
2019/0245223	A1 *	8/2019	Kügele H01M 8/04835
2019/0293313	A1	9/2019	Reeve et al.
2019/0293318	A1	9/2019	Reeve et al.
2019/0338972	A1	11/2019	Schuster et al.
2020/0019129	A1	1/2020	Sircar et al.

OTHER PUBLICATIONS

Johnson Controls York, "Model YK (Style H) R-134a with Optiview™ Control Center for Electro-Mechanical Starter, Solid State Starter and Variable Speed Drive," Jun. 15, 2015, 38 pages.

* cited by examiner



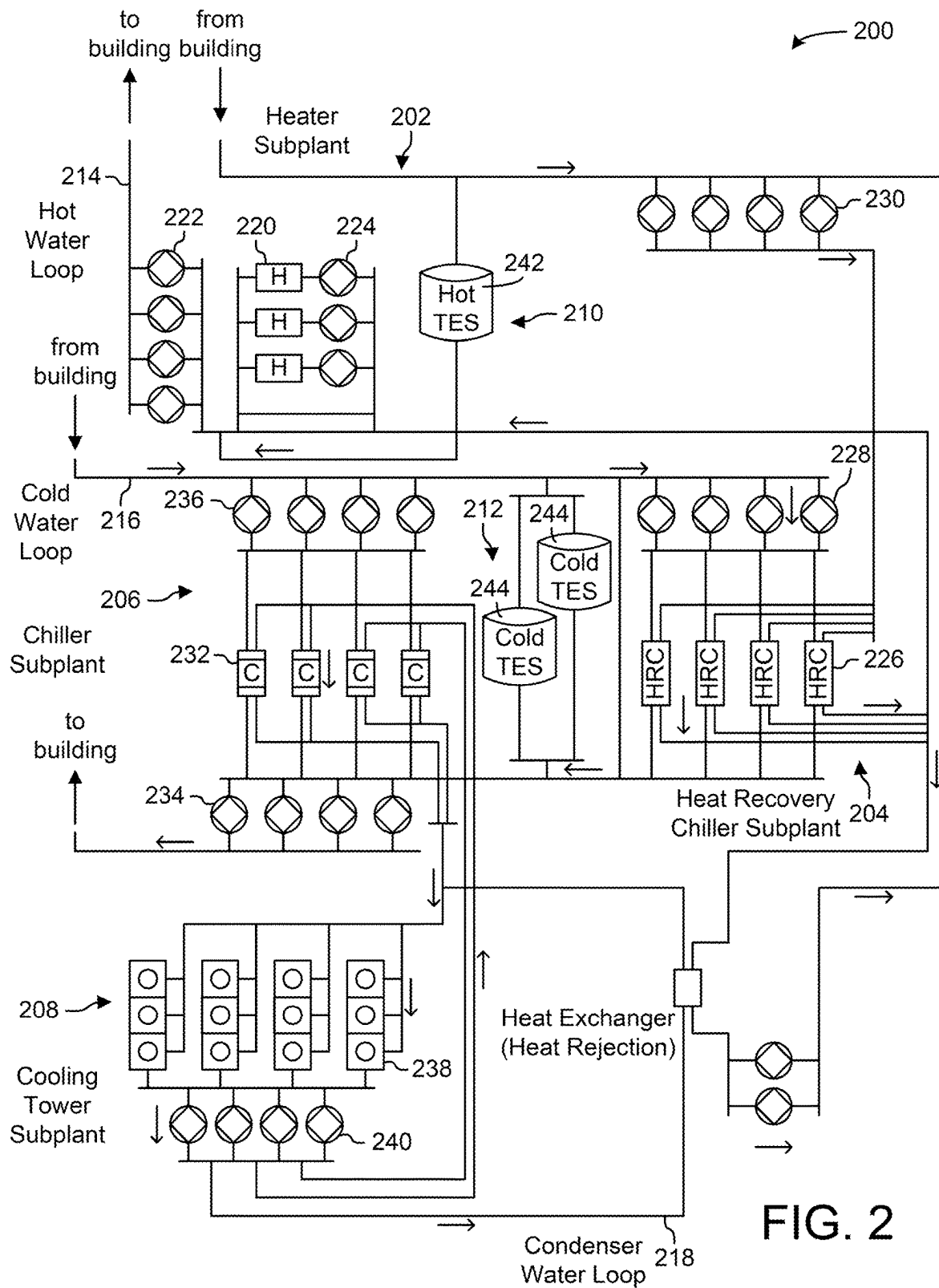


FIG. 2

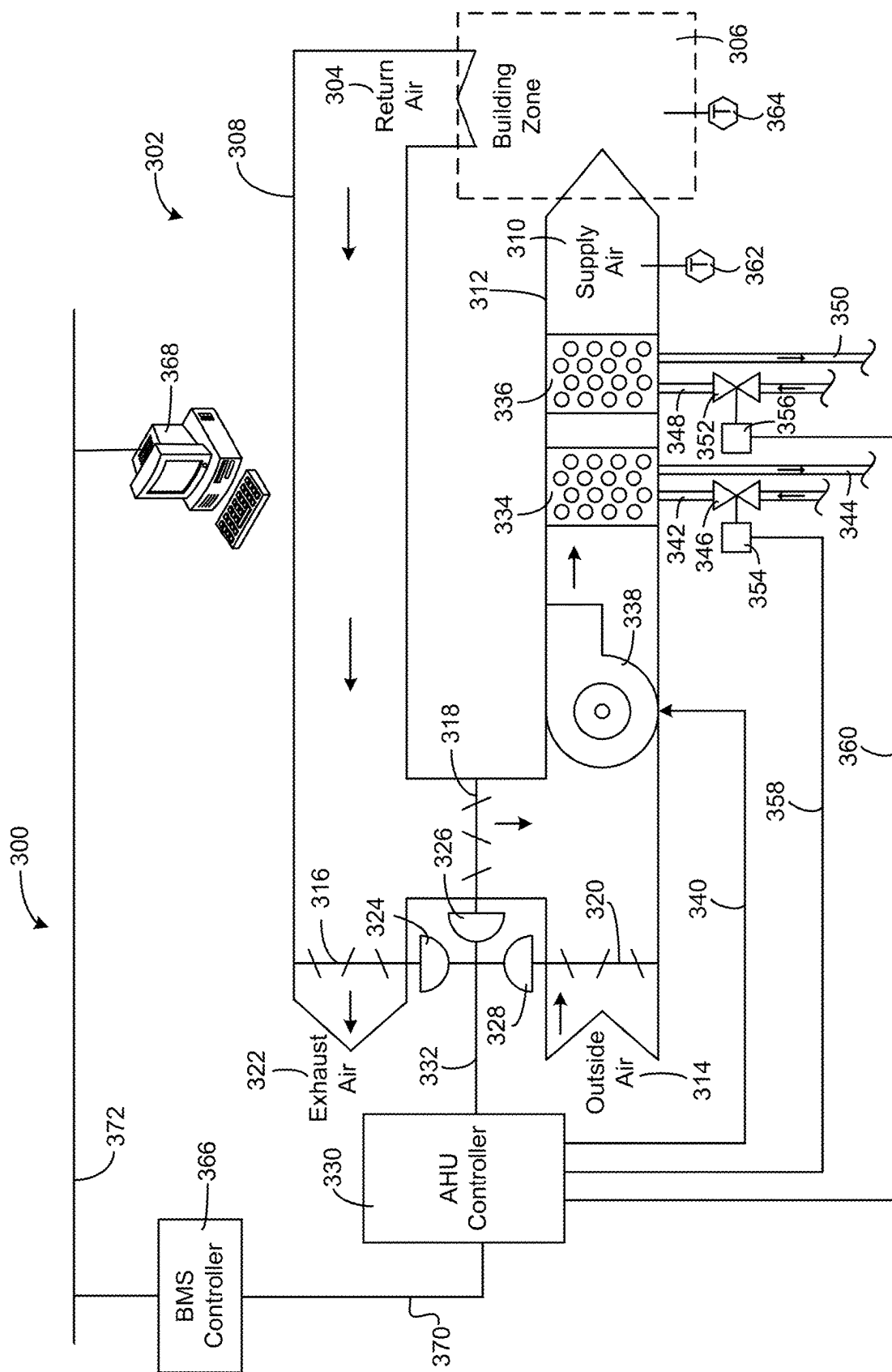


FIG. 3

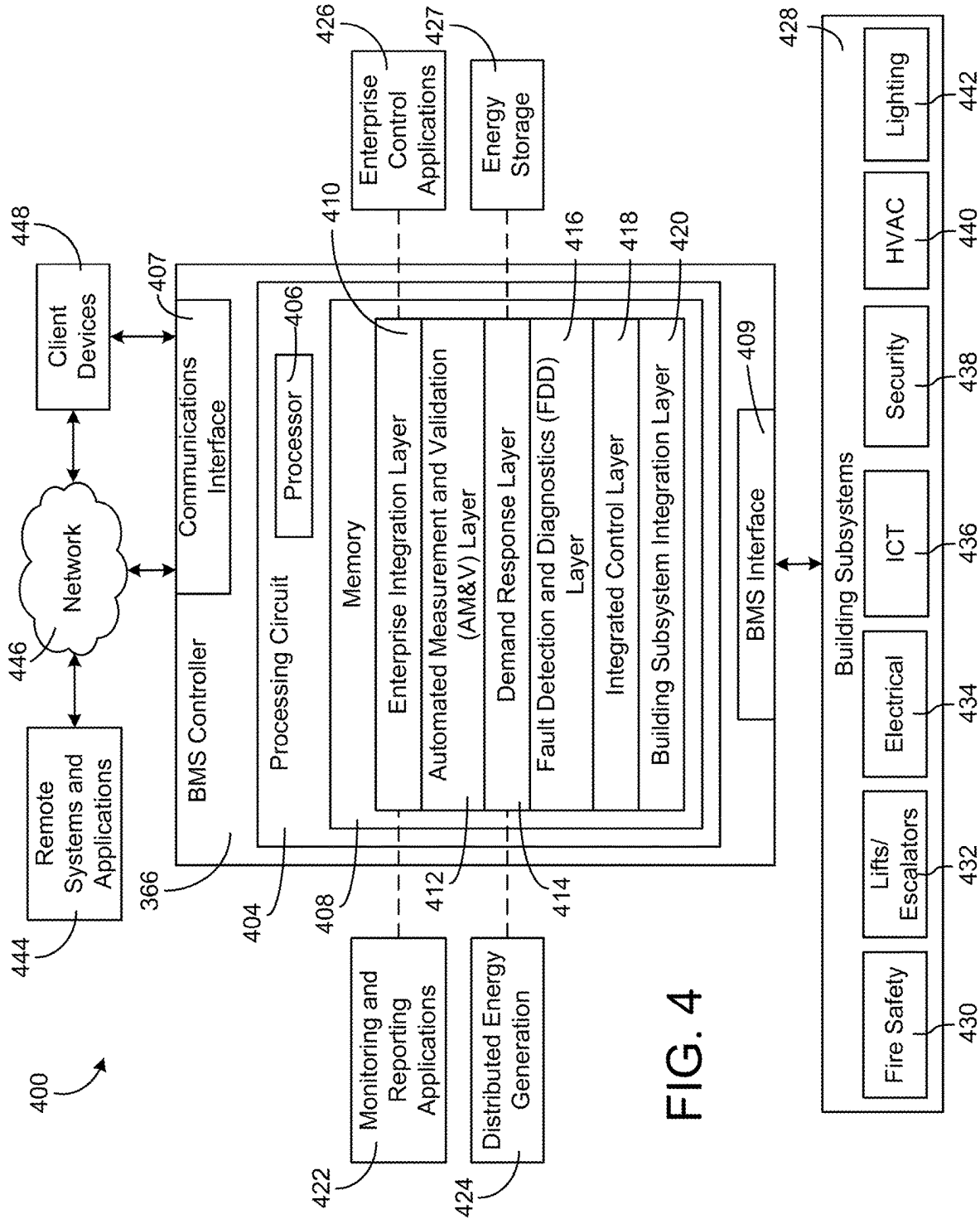


FIG. 4

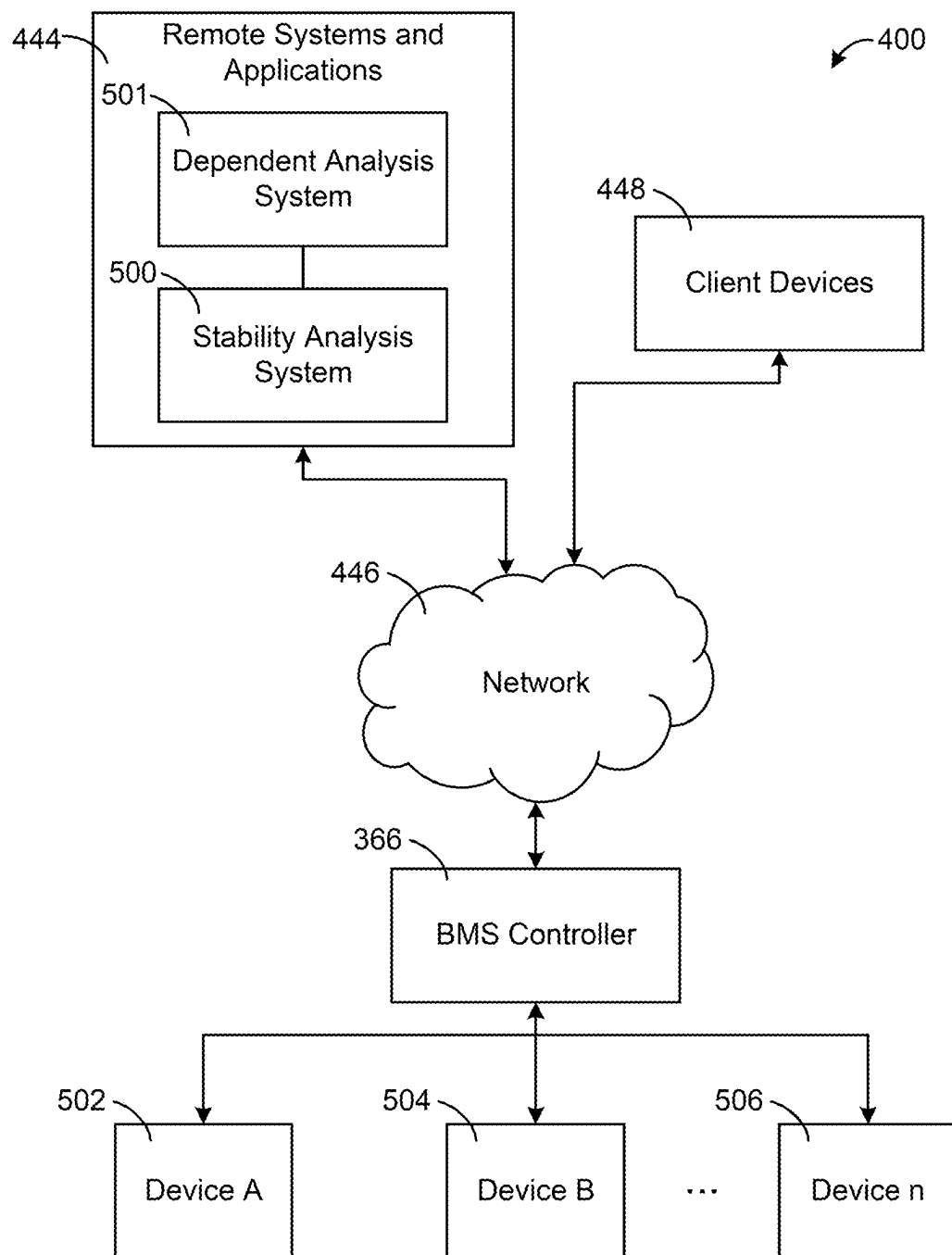


FIG. 5A

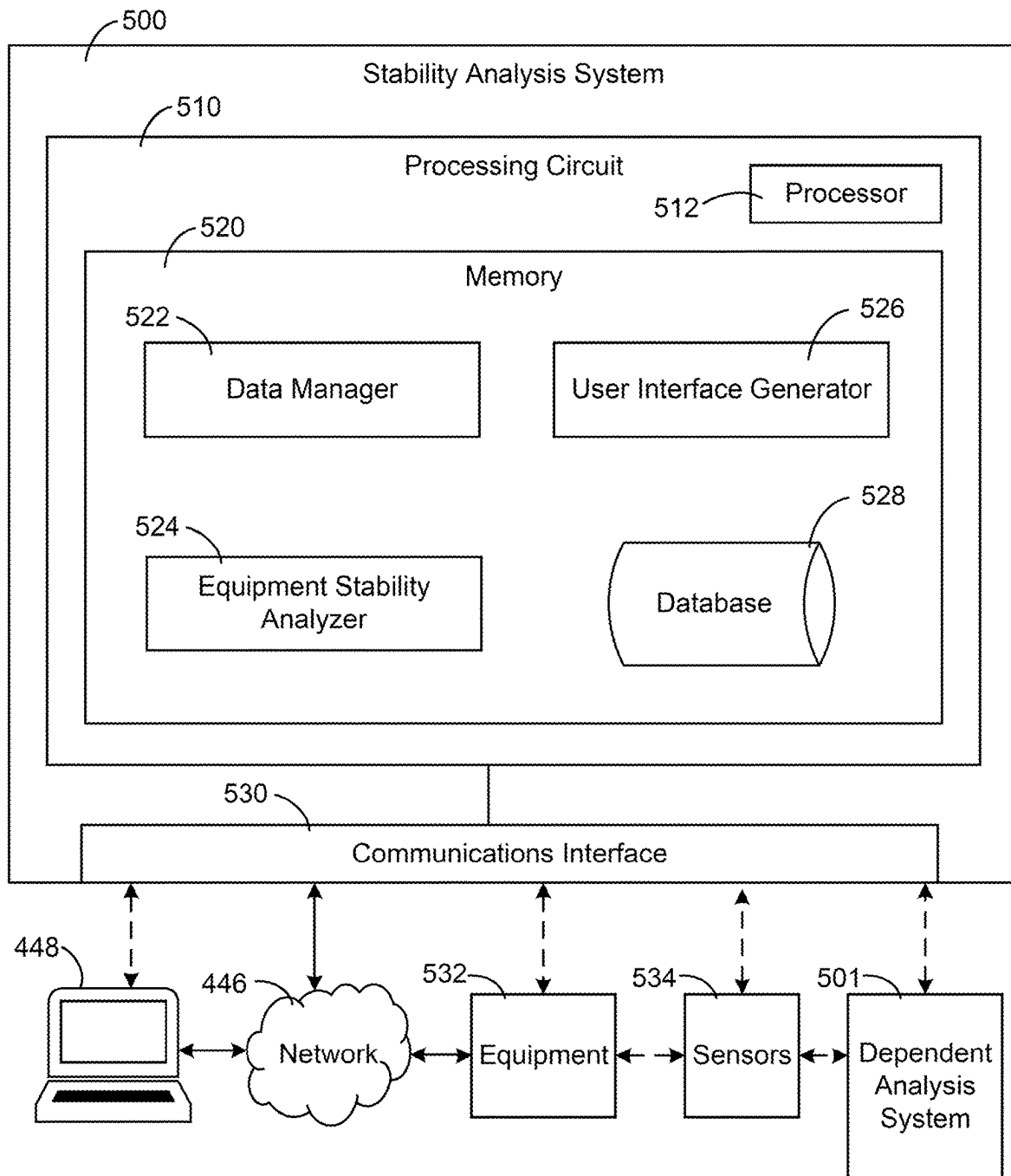


FIG. 5B

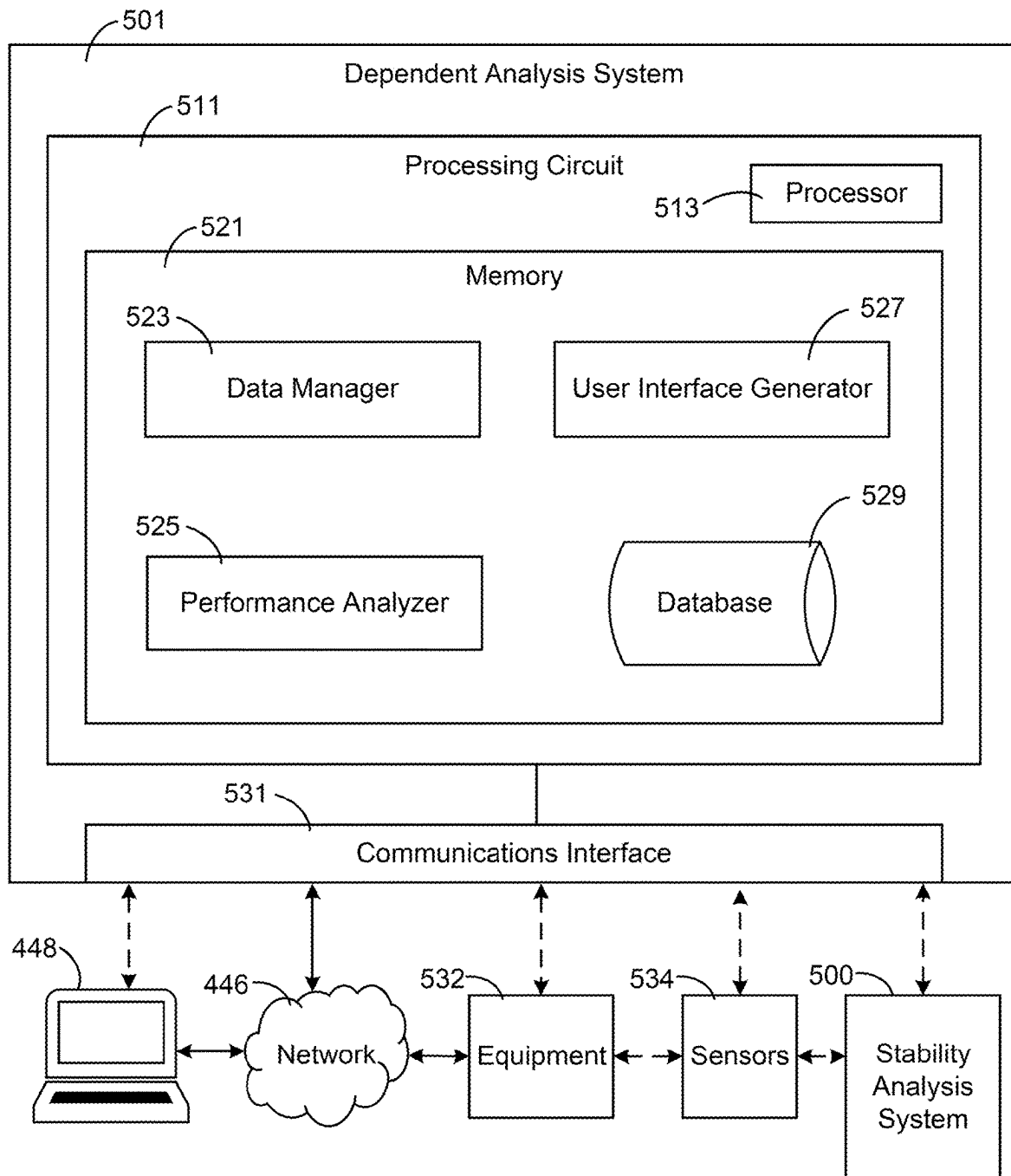


FIG. 5C

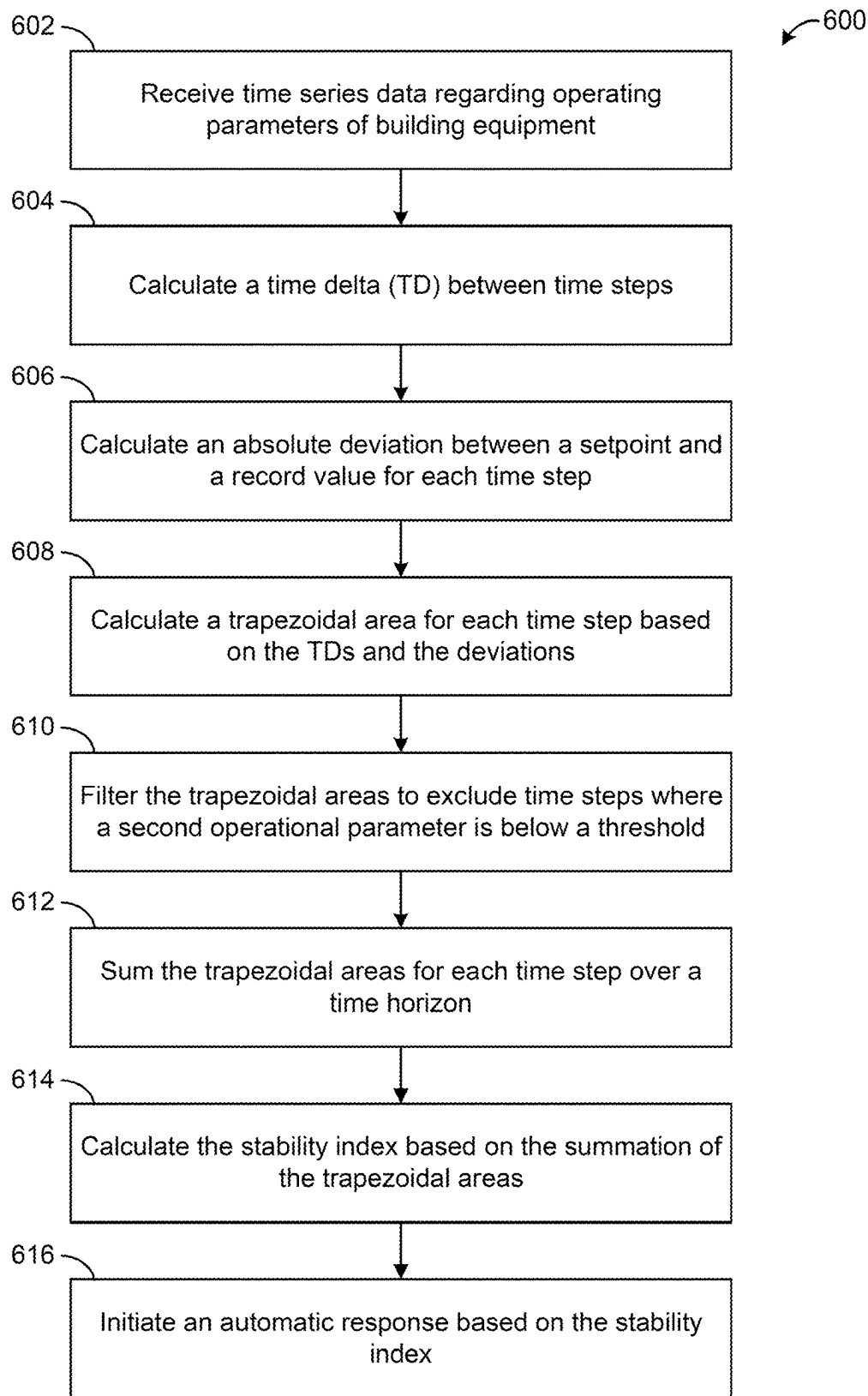


FIG. 6

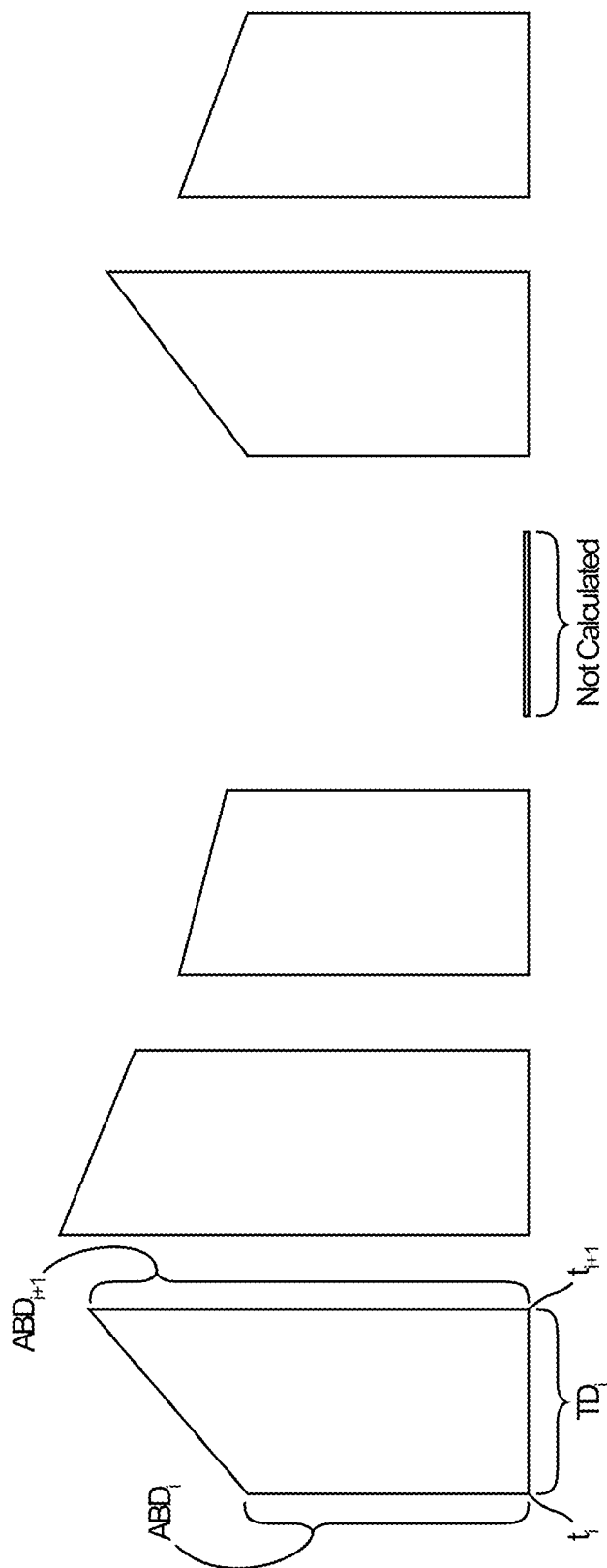


FIG. 7

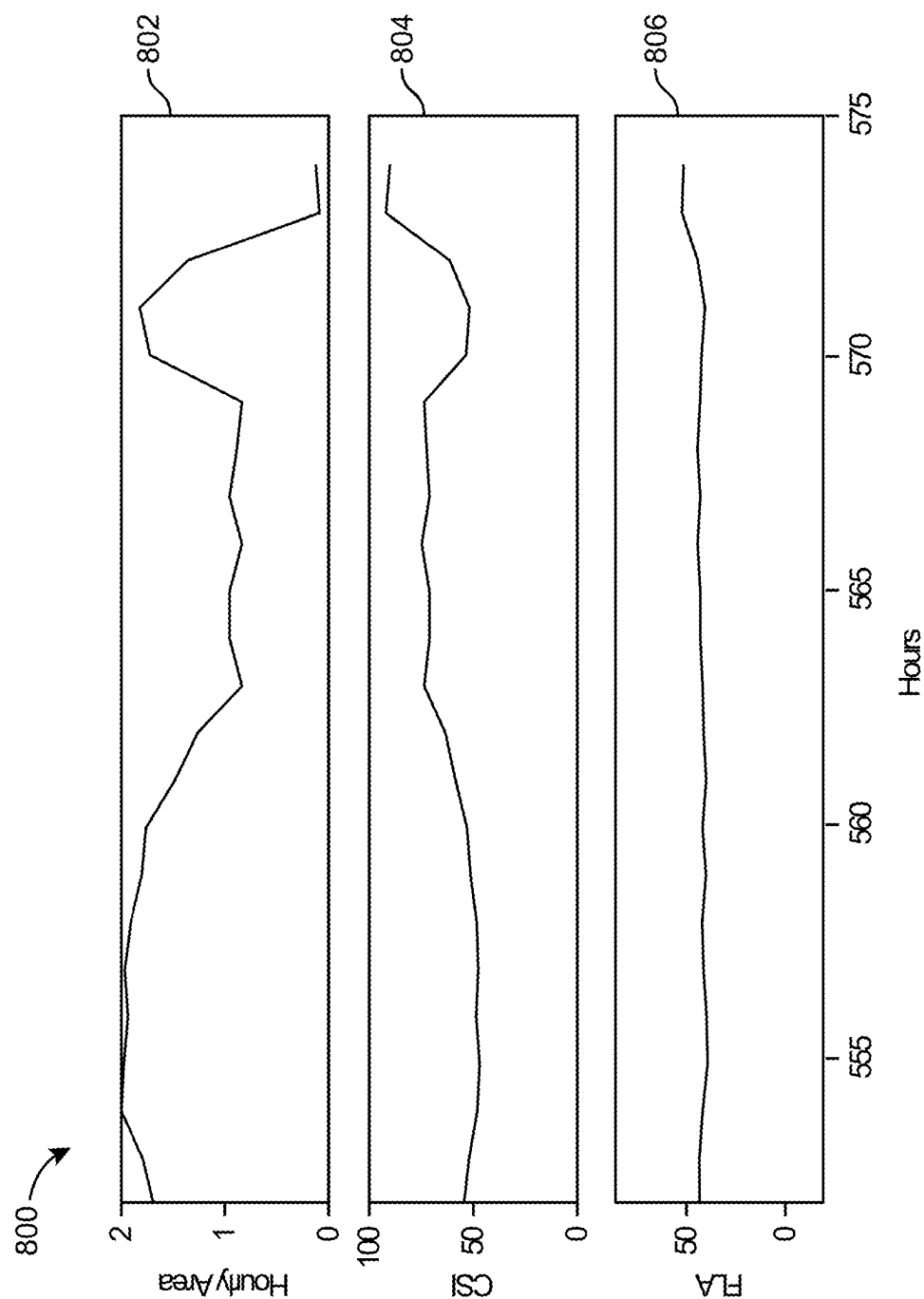


FIG. 8

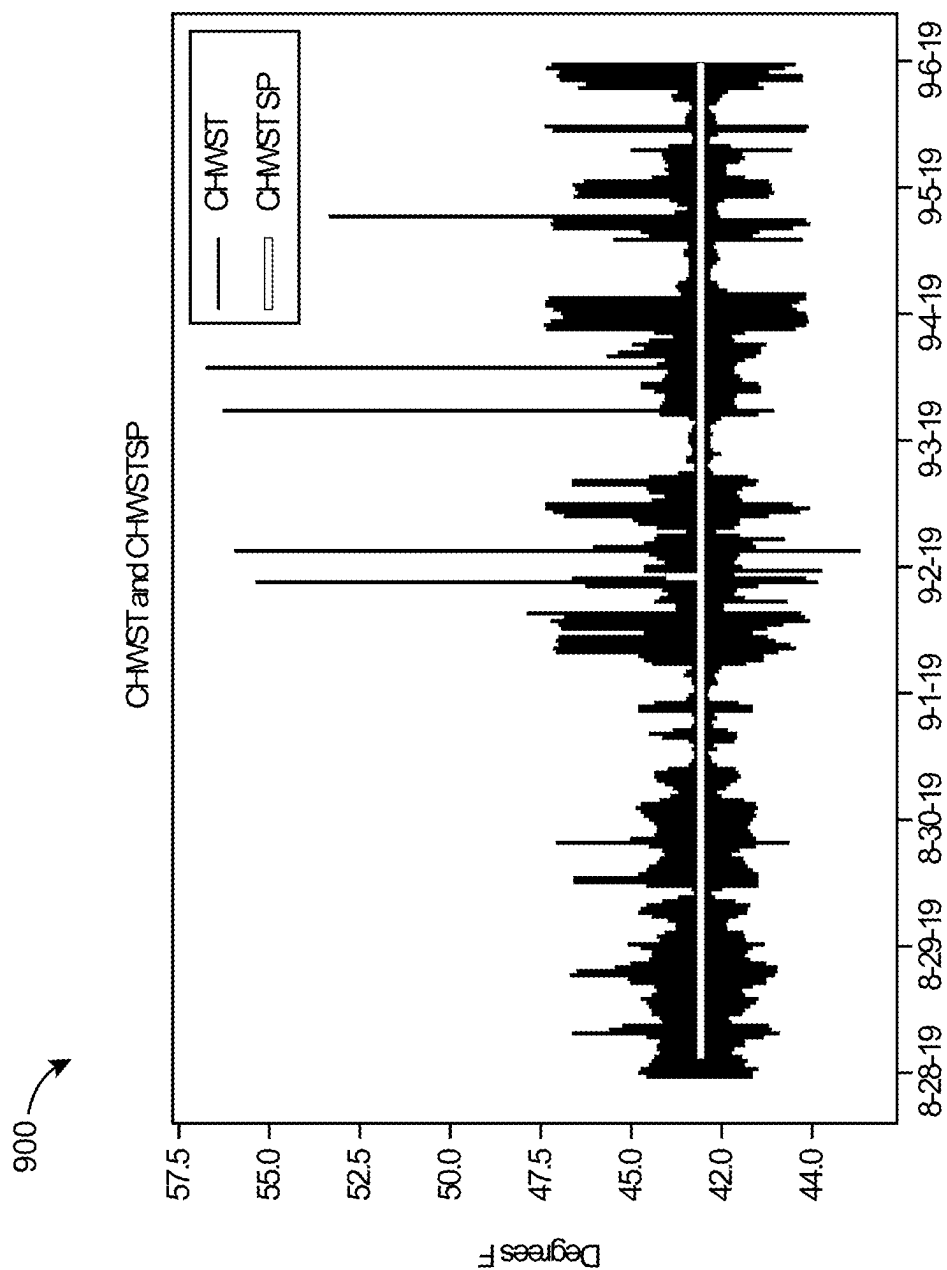


FIG. 9

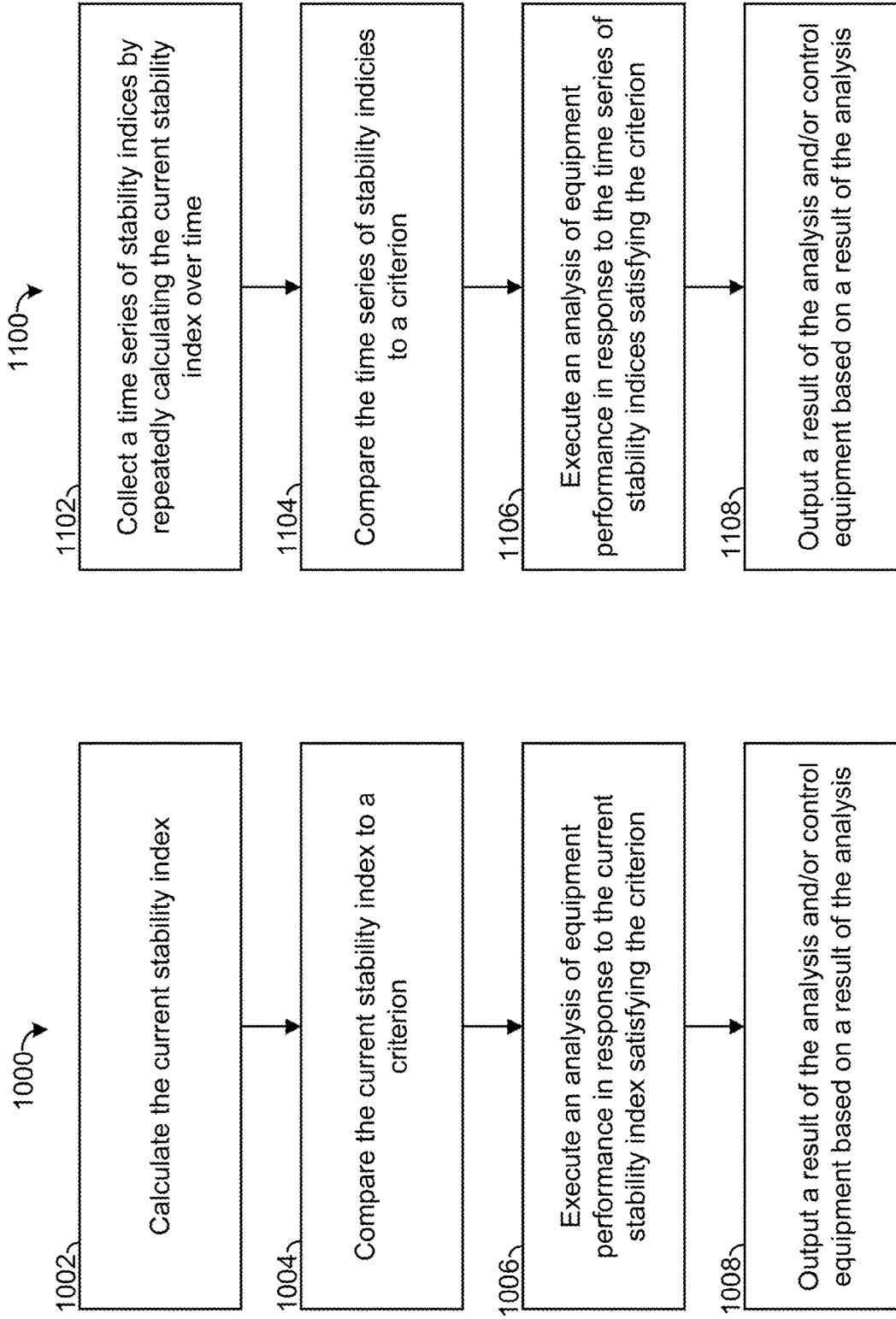


FIG. 10

FIG. 11

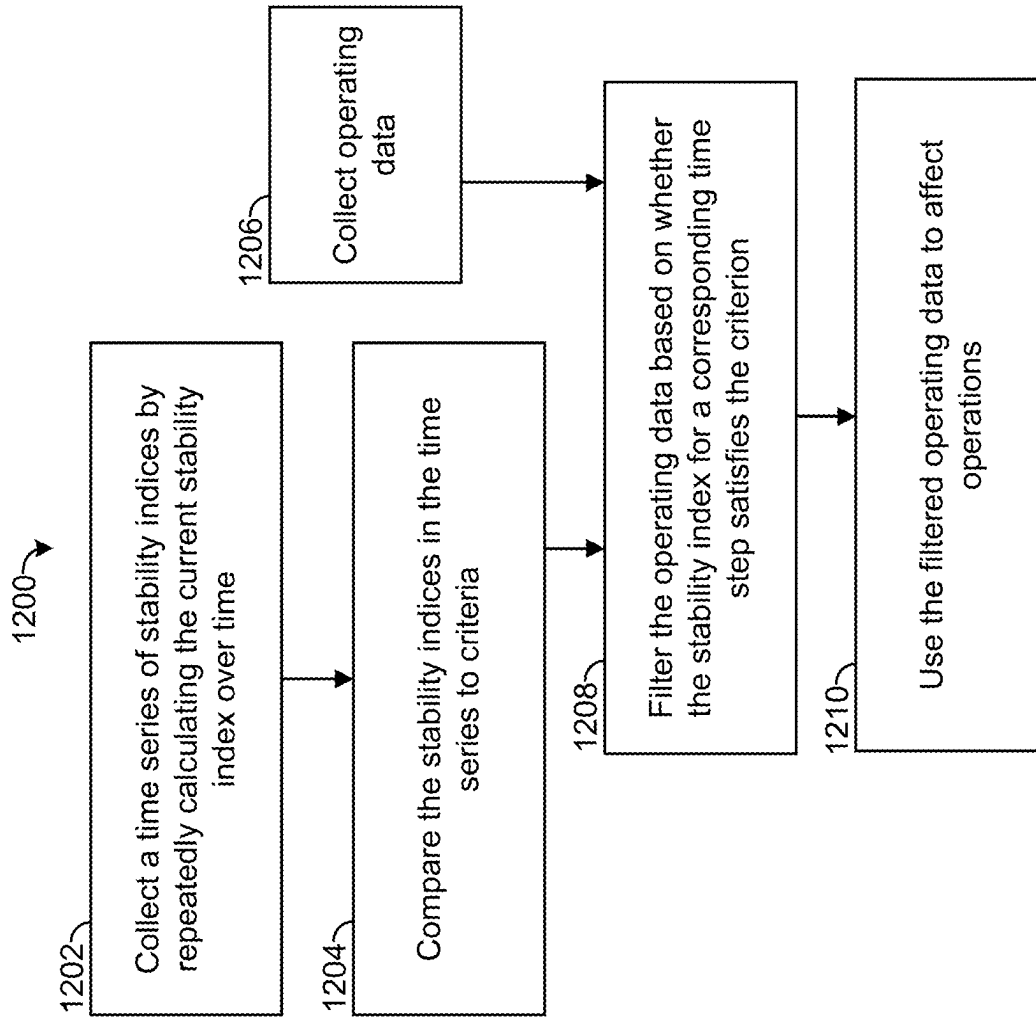


FIG. 12

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PERFORMANCE MONITORING AND CONTROL SYSTEM FOR CONNECTED BUILDING EQUIPMENT WITH STABILITY INDEX

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. application Ser. No. 17/028,704, filed Sep. 22, 2020. This application is also a continuation-in-part of U.S. application Ser. No. 17/400,950, filed Aug. 12, 2021, which is a continuation of U.S. application Ser. No. 16/245,122, filed Jan. 10, 2019 (now U.S. Pat. No. 11,092,954). This application also claims the benefit of and priority to U.S. Provisional Application No. 63/298,720, filed Jan. 12, 2022. The above-cited applications are incorporated by reference herein in their entireties.

BACKGROUND

The present disclosure relates generally to a building management system (BMS), and more particularly, to determining a stability index for building equipment. A BMS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS can include, for example, a HVAC system, a security system, a lighting system, a fire safety system, any other system that is capable of managing building functions or devices, or any combination thereof. Operational data from equipment or devices in a BMS may be monitored to detect and/or prevent problems, and to determine whether the equipment is running as efficiently as possible.

In some cases, a large number of data points or parameters may be required to accurately determine when equipment is experiencing a problem or is running inefficiently. Additionally, some methods for identifying problems or inefficiencies are very specific to a particular type of equipment. In this regard, these methods may need to be modified in order to be applied to other, different types of equipment. Thus, it may be desirable to determine the stability of building equipment using systems and methods that do not require a large number of parameters, and that can be applied to various types of equipment.

SUMMARY

One implementation of the present disclosure is a method for monitoring and controlling building equipment. The method includes obtaining an actual value of a condition affected by operating the building equipment, operating the building equipment using a setpoint for the condition, calculating a stability index based on a timeseries of errors between the actual value and the setpoint, comparing the stability index to a criterion, in response to the stability index satisfying the criterion, executing an action relating to the building equipment, and in response to the stability index not satisfying the criterion, not executing the action.

In some embodiments, the condition comprises at least one of temperature, humidity, pressure, flow rate, light level, or noise level. Calculating the stability index may include calculating a sum of a plurality of trapezoidal areas defined by the timeseries of errors. Executing the action may include performing a fault detection or diagnostic procedure for the building equipment. Executing the action may include modifying a predictive control strategy for the building equip-

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ment and operating the building equipment in accordance with the modified predictive control strategy. The building equipment may include a chiller and executing the action may include calculating a chiller performance index. In some embodiments, the criterion is whether the stability index is greater than a predefined threshold.

Another implementation of the present disclosure is one or more non-transitory computer-readable media storing program instructions that, when executed by one or more processors, cause the one or more processors to perform operations. The operations include obtaining an actual value of a condition affected by operating building equipment, operating the building equipment using a setpoint for the condition, calculating a stability index based on a timeseries of errors between the actual value and the setpoint, comparing the stability index to a criterion, in response to the stability index satisfying the criterion, executing an action relating to the building equipment, and, in response to the stability index not satisfying the criterion, not executing the action.

In some embodiments, the condition comprises at least one of temperature, humidity, pressure, or flowrate. Calculating the stability index may include calculating a sum of a plurality of trapezoidal areas defined by the timeseries of errors. Executing the action may include performing a fault detection or diagnostic procedure for the building equipment. Executing the action may include modifying a predictive control strategy for the building equipment and operating the building equipment in accordance with the modified predictive control strategy. The building equipment may include a chiller and executing the action may include calculating a chiller performance index. In some embodiments, the criterion is whether the stability index is greater than a predefined threshold.

Another implementation of the present disclosure is a building management system. The building management system includes building equipment operable to affect a condition of a building based on a setpoint for the condition, a sensor configured to measure an actual value of the condition, and a computing resource configured to calculate a stability index based on a timeseries of errors between the actual value and the setpoint, compare the stability index to a criterion, in response to the stability index satisfying the criterion, execute an action relating to the building equipment, and in response to the stability index not satisfying the criterion, not execute the action.

In some embodiments, the condition is temperature, humidity, pressure or flowrate. The computing resource may be configured to calculate the stability index by calculating a sum of a plurality of trapezoidal areas defined by the timeseries of errors. The computing resource may be configured to execute the action by performing a fault detection or diagnostic procedure for the building equipment. The computing resource may be configured to execute the action by modifying a predictive control strategy for the building equipment and operating the building equipment in accordance with the modified predictive control strategy. In some embodiments, the computing resource is configured to compare the stability index to the criterion by determining the stability index is less than a predefined threshold.

BRIEF DESCRIPTION OF THE DRAWINGS

Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference

characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

FIG. 1 is a drawing of a building equipped with a HVAC system, according to some embodiments.

FIG. 2 is a block diagram of a waterside system that may be used in conjunction with the building of FIG. 1, according to some embodiments.

FIG. 3 is a block diagram of an airside system that may be used in conjunction with the building of FIG. 1, according to some embodiments.

FIG. 4 is a block diagram of a building management system (BMS) that may be used to monitor and/or control the building of FIG. 1, according to some embodiments.

FIG. 5A is an additional block diagram of the BMS of FIG. 4 illustrating a communications network, according to some embodiments.

FIG. 5B is a block diagram of a stability analysis system, according to some embodiments.

FIG. 5C is a block diagram of a dependent analysis system, according to some embodiments.

FIG. 6 is an example process for determining a stability index value for equipment, according to some embodiments.

FIG. 7 is an example graph illustrating trapezoidal areas over a plurality of time periods, according to some embodiments.

FIG. 8 is an example interface for presenting a stability index and various other parameters associated with equipment operations, according to some embodiments.

FIG. 9 is an example interface presenting chilled water temperatures over time, according to some embodiments.

FIG. 10 is a flowchart of a process for using a stability index value, according to some embodiments.

FIG. 11 is a flowchart of a process for using a time series for using a time series of stability index values, according to some embodiments.

FIG. 12 is a flowchart of another process for using a time series for using a time series of stability index values, according to some embodiments.

DETAILED DESCRIPTION

Overview

Referring generally to the FIGURES, a stability analysis system is shown, according to some embodiments. At a high level, the stability analysis system may be configured to monitor operational data from building equipment and generate a stability index value that indicates how closely the building equipment is maintaining setpoints. The stability index value may be a quantitative indicator of equipment performance, and in some cases can be utilized to diagnose problems with the building equipment and/or to determine the overall equipment health. The stability index may also indicate that the building equipment requires maintenance or service in order to correct or prevent problems, which may lead to unnecessary equipment down time, costly repairs, equipment failure, etc.

The stability analysis system can include a controller or other central processing unit that monitors and/or receives operational data from building equipment and/or a database. The operational data can be applied to an algorithm or a model that calculates the stability index. Based on the stability index, any number of automated responses may be initiated. For example, if the stability index indicates a problem with the building equipment, maintenance or service can be automatically scheduled for the affected equip-

ment. The stability index may also be used to determine the health of the equipment or system, and can be used to generate a health dashboard (i.e., graphical user interface). Building With Building Systems

Referring now to FIGS. 1-4, an exemplary building management system (BMS) and HVAC system in which the systems and methods of the present disclosure can be implemented are shown, according to some embodiments. Referring particularly to FIG. 1, a perspective view of a building 10 is shown. Building 10 is served by a BMS. As discussed above, a BMS is generally a system of devices configured to control, monitor, and manage equipment in or around a building or building area, and a BMS can include, for example, a HVAC system, a security system, a lighting system, a fire safety system, any other system that is capable of managing building functions or devices, or any combination thereof.

The BMS that serves building 10 includes an HVAC system 100. HVAC system 100 can include a plurality of HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage, etc.) configured to provide heating, cooling, ventilation, or other services for building 10. For example, HVAC system 100 is shown to include a waterside system 120 and an airside system 130. Waterside system 120 can provide a heated or chilled fluid to an air handling unit of airside system 130. Airside system 130 can use the heated or chilled fluid to heat or cool an airflow provided to building 10. An exemplary waterside system and airside system which can be used in HVAC system 100 are described in greater detail with reference to FIGS. 2-3.

HVAC system 100 is shown to include a chiller 102, a boiler 104, and a rooftop air handling unit (AHU) 106. Waterside system 120 can use boiler 104 and chiller 102 to heat or cool a working fluid (e.g., water, glycol, etc.) and can circulate the working fluid to AHU 106. In various embodiments, the HVAC devices of waterside system 120 can be located in or around building 10 (as shown in FIG. 1) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant, etc.). The working fluid can be heated in boiler 104 or cooled in chiller 102, depending on whether heating or cooling is required in building 10. Boiler 104 can add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. Chiller 102 can place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from chiller 102 and/or boiler 104 can be transported to AHU 106 via piping 108.

AHU 106 can place the working fluid in a heat exchange relationship with an airflow passing through AHU 106 (e.g., via one or more stages of cooling coils and/or heating coils). The airflow can be, for example, outside air, return air from within building 10, or a combination of both. AHU 106 can transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow. For example, AHU 106 can include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid can then return to chiller 102 or boiler 104 via piping 110.

Airside system 130 can deliver the airflow supplied by AHU 106 (i.e., the supply airflow) to building 10 via air supply ducts 112 and can provide return air from building 10 to AHU 106 via air return ducts 114. In some embodiments, airside system 130 includes multiple variable air volume (VAV) units 116. For example, airside system 130 is shown

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to include a separate VAV unit **116** on each floor or zone of building **10**. VAV units **116** can include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of building **10**. In other embodiments, airside system **130** delivers the supply airflow into one or more zones of building **10** (e.g., via supply ducts **112**) without using intermediate VAV units **116** or other flow control elements. AHU **106** can include various sensors (e.g., temperature sensors, pressure sensors, etc.) configured to measure attributes of the supply airflow. AHU **106** can receive input from sensors located within AHU **106** and/or within the building zone and can adjust the flow rate, temperature, or other attributes of the supply airflow through AHU **106** to achieve setpoint conditions for the building zone.

In FIG. 2, waterside system **200** is shown as a central plant having a plurality of subplants **202-212**. Subplants **202-212** are shown to include a heater subplant **202**, a heat recovery chiller subplant **204**, a chiller subplant **206**, a cooling tower subplant **208**, a hot thermal energy storage (TES) subplant **210**, and a cold thermal energy storage (TES) subplant **212**. Subplants **202-212** consume resources (e.g., water, natural gas, electricity, etc.) from utilities to serve the thermal energy loads (e.g., hot water, cold water, heating, cooling, etc.) of a building or campus. For example, heater subplant **202** may be configured to heat water in a hot water loop **214** that circulates the hot water between heater subplant **202** and building **10**. Chiller subplant **206** may be configured to chill water in a cold water loop **216** that circulates the cold water between chiller subplant **206** building **10**. Heat recovery chiller subplant **204** may be configured to transfer heat from cold water loop **216** to hot water loop **214** to provide additional heating for the hot water and additional cooling for the cold water. Condenser water loop **218** may absorb heat from the cold water in chiller subplant **206** and reject the absorbed heat in cooling tower subplant **208** or transfer the absorbed heat to hot water loop **214**. Hot TES subplant **210** and cold TES subplant **212** may store hot and cold thermal energy, respectively, for subsequent use.

Hot water loop **214** and cold water loop **216** may deliver the heated and/or chilled water to air handlers located on the rooftop of building **10** (e.g., AHU **106**) or to individual floors or zones of building **10** (e.g., VAV units **116**). The air handlers push air past heat exchangers (e.g., heating coils or cooling coils) through which the water flows to provide heating or cooling for the air. The heated or cooled air may be delivered to individual zones of building **10** to serve the thermal energy loads of building **10**. The water then returns to subplants **202-212** to receive further heating or cooling.

Although subplants **202-212** are shown and described as heating and cooling water for circulation to a building, it is understood that any other type of working fluid (e.g., glycol, CO₂, etc.) may be used in place of or in addition to water to serve the thermal energy loads. In other embodiments, subplants **202-212** may provide heating and/or cooling directly to the building or campus without requiring an intermediate heat transfer fluid. These and other variations to waterside system **200** are within the teachings of the present invention.

Each of subplants **202-212** may include a variety of equipment configured to facilitate the functions of the subplant. For example, heater subplant **202** is shown to include a plurality of heating elements **220** (e.g., boilers, electric heaters, etc.) configured to add heat to the hot water in hot water loop **214**. Heater subplant **202** is also shown to include several pumps **222** and **224** configured to circulate the hot water in hot water loop **214** and to control the flow rate of

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the hot water through individual heating elements **220**. Chiller subplant **206** is shown to include a plurality of chillers **232** configured to remove heat from the cold water in cold water loop **216**. Chiller subplant **206** is also shown to include several pumps **234** and **236** configured to circulate the cold water in cold water loop **216** and to control the flow rate of the cold water through individual chillers **232**.

Heat recovery chiller subplant **204** is shown to include a plurality of heat recovery heat exchangers **226** (e.g., refrigeration circuits) configured to transfer heat from cold water loop **216** to hot water loop **214**. Heat recovery chiller subplant **204** is also shown to include several pumps **228** and **230** configured to circulate the hot water and/or cold water through heat recovery heat exchangers **226** and to control the flow rate of the water through individual heat recovery heat exchangers **226**. Cooling tower subplant **208** is shown to include a plurality of cooling towers **238** configured to remove heat from the condenser water in condenser water loop **218**. Cooling tower subplant **208** is also shown to include several pumps **240** configured to circulate the condenser water in condenser water loop **218** and to control the flow rate of the condenser water through individual cooling towers **238**.

Hot TES subplant **210** is shown to include a hot TES tank **242** configured to store the hot water for later use. Hot TES subplant **210** may also include one or more pumps or valves configured to control the flow rate of the hot water into or out of hot TES tank **242**. Cold TES subplant **212** is shown to include cold TES tanks **244** configured to store the cold water for later use. Cold TES subplant **212** may also include one or more pumps or valves configured to control the flow rate of the cold water into or out of cold TES tanks **244**.

In some embodiments, one or more of the pumps in waterside system **200** (e.g., pumps **222**, **224**, **228**, **230**, **234**, **236**, and/or **240**) or pipelines in waterside system **200** include an isolation valve associated therewith. Isolation valves may be integrated with the pumps or positioned upstream or downstream of the pumps to control the fluid flows in waterside system **200**. In various embodiments, waterside system **200** may include more, fewer, or different types of devices and/or subplants based on the particular configuration of waterside system **200** and the types of loads served by waterside system **200**.

Referring now to FIG. 3, a block diagram of an airside system **300** is shown, according to some embodiments. In various embodiments, airside system **300** may supplement or replace airside system **130** in HVAC system **100** or may be implemented separate from HVAC system **100**. When implemented in HVAC system **100**, airside system **300** may include a subset of the HVAC devices in HVAC system **100** (e.g., AHU **106**, VAV units **116**, ducts **112-114**, fans, dampers, etc.) and may be located in or around building **10**. Airside system **300** may operate to heat or cool an airflow provided to building **10** using a heated or chilled fluid provided by waterside system **200**.

In FIG. 3, airside system **300** is shown to include an economizer-type air handling unit (AHU) **302**. Economizer-type AHUs vary the amount of outside air and return air used by the air handling unit for heating or cooling. For example, AHU **302** may receive return air **304** from building zone **306** via return air duct **308** and may deliver supply air **310** to building zone **306** via supply air duct **312**. In some embodiments, AHU **302** is a rooftop unit located on the roof of building **10** (e.g., AHU **106** as shown in FIG. 1) or otherwise positioned to receive both return air **304** and outside air **314**. AHU **302** may be configured to operate exhaust air damper **316**, mixing damper **318**, and outside air damper **320** to

control an amount of outside air **314** and return air **304** that combine to form supply air **310**. Any return air **304** that does not pass through mixing damper **318** may be exhausted from AHU **302** through exhaust damper **316** as exhaust air **322**.

Each of dampers **316-320** may be operated by an actuator. For example, exhaust air damper **316** may be operated by actuator **324**, mixing damper **318** may be operated by actuator **326**, and outside air damper **320** may be operated by actuator **328**. Actuators **324-328** may communicate with an AHU controller **330** via a communications link **332**. Actuators **324-328** may receive control signals from AHU controller **330** and may provide feedback signals to AHU controller **330**. Feedback signals may include, for example, an indication of a current actuator or damper position, an amount of torque or force exerted by the actuator, diagnostic information (e.g., results of diagnostic tests performed by actuators **324-328**), status information, commissioning information, configuration settings, calibration data, and/or other types of information or data that may be collected, stored, or used by actuators **324-328**. AHU controller **330** may be an economizer controller configured to use one or more control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control actuators **324-328**.

Still referring to FIG. 3, AHU **302** is shown to include a cooling coil **334**, a heating coil **336**, and a fan **338** positioned within supply air duct **312**. Fan **338** may be configured to force supply air **310** through cooling coil **334** and/or heating coil **336** and provide supply air **310** to building zone **306**. AHU controller **330** may communicate with fan **338** via communications link **340** to control a flow rate of supply air **310**. In some embodiments, AHU controller **330** controls an amount of heating or cooling applied to supply air **310** by modulating a speed of fan **338**.

Cooling coil **334** may receive a chilled fluid from waterside system **200** (e.g., from cold water loop **216**) via piping **342** and may return the chilled fluid to waterside system **200** via piping **344**. Valve **346** may be positioned along piping **342** or piping **344** to control a flow rate of the chilled fluid through cooling coil **334**. In some embodiments, cooling coil **334** includes multiple stages of cooling coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BMS controller **366**, etc.) to modulate an amount of cooling applied to supply air **310**.

Heating coil **336** may receive a heated fluid from waterside system **200** (e.g., from hot water loop **214**) via piping **348** and may return the heated fluid to waterside system **200** via piping **350**. Valve **352** may be positioned along piping **348** or piping **350** to control a flow rate of the heated fluid through heating coil **336**. In some embodiments, heating coil **336** includes multiple stages of heating coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BMS controller **366**, etc.) to modulate an amount of heating applied to supply air **310**.

Each of valves **346** and **352** may be controlled by an actuator. For example, valve **346** may be controlled by actuator **354** and valve **352** may be controlled by actuator **356**. Actuators **354-356** may communicate with AHU controller **330** via communications links **358-360**. Actuators **354-356** may receive control signals from AHU controller **330** and may provide feedback signals to controller **330**. In some embodiments, AHU controller **330** receives a measurement of the supply air temperature from a temperature sensor **362** positioned in supply air duct **312** (e.g., down-

stream of cooling coil **334** and/or heating coil **336**). AHU controller **330** may also receive a measurement of the temperature of building zone **306** from a temperature sensor **364** located in building zone **306**.

In some embodiments, AHU controller **330** operates valves **346** and **352** via actuators **354-356** to modulate an amount of heating or cooling provided to supply air **310** (e.g., to achieve a setpoint temperature for supply air **310** or to maintain the temperature of supply air **310** within a setpoint temperature range). The positions of valves **346** and **352** affect the amount of heating or cooling provided to supply air **310** by cooling coil **334** or heating coil **336** and may correlate with the amount of energy consumed to achieve a desired supply air temperature. AHU controller **330** may control the temperature of supply air **310** and/or building zone **306** by activating or deactivating coils **334-336**, adjusting a speed of fan **338**, or a combination of both.

Still referring to FIG. 3, airside system **300** is shown to include a building automation system (BMS) controller **366** and a client device **368**. BMS controller **366** may include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers, etc.) that serve as system level controllers, application or data servers, head nodes, or master controllers for airside system **300**, waterside system **200**, HVAC system **100**, and/or other controllable systems that serve building **10**. BMS controller **366** may communicate with multiple downstream building systems or subsystems (e.g., HVAC system **100**, a security system, a lighting system, waterside system **200**, etc.) via a communications link **370** according to like or disparate protocols (e.g., LON, BACnet, etc.). In various embodiments, AHU controller **330** and BMS controller **366** may be separate (as shown in FIG. 3) or integrated. In an integrated implementation, AHU controller **330** may be a software module configured for execution by a processor of BMS controller **366**.

In some embodiments, AHU controller **330** receives information from BMS controller **366** (e.g., commands, setpoints, operating boundaries, etc.) and provides information to BMS controller **366** (e.g., temperature measurements, valve or actuator positions, operating statuses, diagnostics, etc.). For example, AHU controller **330** may provide BMS controller **366** with temperature measurements from temperature sensors **362-364**, equipment on/off states, equipment operating capacities, and/or any other information that can be used by BMS controller **366** to monitor or control a variable state or condition within building zone **306**.

Client device **368** may include one or more human-machine interfaces or client interfaces (e.g., graphical user interfaces, reporting interfaces, text-based computer interfaces, client-facing web services, web servers that provide pages to web clients, etc.) for controlling, viewing, or otherwise interacting with HVAC system **100**, its subsystems, and/or devices. Client device **368** may be a computer workstation, a client terminal, a remote or local interface, or any other type of user interface device. Client device **368** may be a stationary terminal or a mobile device. For example, client device **368** may be a desktop computer, a computer server with a user interface, a laptop computer, a tablet, a smartphone, a PDA, or any other type of mobile or non-mobile device. Client device **368** may communicate with BMS controller **366** and/or AHU controller **330** via communications link **372**.

Referring now to FIG. 4, a block diagram of a building management system (BMS) **400** is shown, according to some embodiments. BMS **400** may be implemented in building **10** to automatically monitor and control various building functions. BMS **400** is shown to include BMS

controller **366** and a plurality of building subsystems **428**. Building subsystems **428** are shown to include a building electrical subsystem **434**, an information communication technology (ICT) subsystem **436**, a security subsystem **438**, a HVAC subsystem **440**, a lighting subsystem **442**, a lift/escalators subsystem **432**, and a fire safety subsystem **430**. In various embodiments, building subsystems **428** can include fewer, additional, or alternative subsystems. For example, building subsystems **428** may also or alternatively include a refrigeration subsystem, an advertising or signage subsystem, a cooking subsystem, a vending subsystem, a printer or copy service subsystem, or any other type of building subsystem that uses controllable equipment and/or sensors to monitor or control building **10**. In some embodiments, building subsystems **428** include waterside system **200** and/or airside system **300**, as described with reference to FIGS. 2-3.

Each of building subsystems **428** may include any number of devices, controllers, and connections for completing its individual functions and control activities. HVAC subsystem **440** may include many of the same components as HVAC system **100**, as described with reference to FIGS. 1-3. For example, HVAC subsystem **440** may include a chiller, a boiler, any number of air handling units, economizers, field controllers, supervisory controllers, actuators, temperature sensors, and other devices for controlling the temperature, humidity, airflow, or other variable conditions within building **10**. Lighting subsystem **442** may include any number of light fixtures, ballasts, lighting sensors, dimmers, or other devices configured to controllably adjust the amount of light provided to a building space. Security subsystem **438** may include occupancy sensors, video surveillance cameras, digital video recorders, video processing servers, intrusion detection devices, access control devices and servers, or other security-related devices.

Still referring to FIG. 4, BMS controller **366** is shown to include a communications interface **407** and a BMS interface **409**. Interface **407** may facilitate communications between BMS controller **366** and external applications (e.g., monitoring and reporting applications **422**, enterprise control applications **426**, remote systems and applications **444**, applications residing on client devices **448**, etc.) for allowing user control, monitoring, and adjustment to BMS controller **366** and/or subsystems **428**. Interface **407** may also facilitate communications between BMS controller **366** and client devices **448**. BMS interface **409** may facilitate communications between BMS controller **366** and building subsystems **428** (e.g., HVAC, lighting security, lifts, power distribution, business, etc.).

Interfaces **407**, **409** can be or include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with building subsystems **428** or other external systems or devices. In various embodiments, communications via interfaces **407**, **409** may be direct (e.g., local wired or wireless communications) or via a communications network **446** (e.g., a WAN, the Internet, a cellular network, etc.). For example, interfaces **407**, **409** can include an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. In another example, interfaces **407**, **409** can include a WiFi transceiver for communicating via a wireless communications network. In another example, one or both of interfaces **407**, **409** may include cellular or mobile phone communications transceivers. In one embodiment, communications interface **407** is a power line communications interface and BMS interface **409** is an Ethernet interface. In

other embodiments, both communications interface **407** and BMS interface **409** are Ethernet interfaces or are the same Ethernet interface.

Still referring to FIG. 4, BMS controller **366** is shown to include a processing circuit **404** including a processor **406** and memory **408**. Processing circuit **404** may be communicably connected to BMS interface **409** and/or communications interface **407** such that processing circuit **404** and the various components thereof can send and receive data via interfaces **407**, **409**. Processor **406** can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

Memory **408** (e.g., memory, memory unit, storage device, etc.) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **408** may be or include volatile memory or non-volatile memory. Memory **408** may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to an exemplary embodiment, memory **408** is communicably connected to processor **406** via processing circuit **404** and includes computer code for executing (e.g., by processing circuit **404** and/or processor **406**) one or more processes described herein.

In some embodiments, BMS controller **366** is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments BMS controller **366** may be distributed across multiple servers or computers (e.g., that can exist in distributed locations). Further, while FIG. 4 shows applications **422** and **426** as existing outside of BMS controller **366**, in some embodiments, applications **422** and **426** may be hosted within BMS controller **366** (e.g., within memory **408**).

Still referring to FIG. 4, memory **408** is shown to include an enterprise integration layer **410**, an automated measurement and validation (AM&V) layer **412**, a demand response (DR) layer **414**, a fault detection and diagnostics (FDD) layer **416**, an integrated control layer **418**, and a building subsystem integration later **420**. Layers **410-420** may be configured to receive inputs from building subsystems **428** and other data sources, determine optimal control actions for building subsystems **428** based on the inputs, generate control signals based on the optimal control actions, and provide the generated control signals to building subsystems **428**. The following paragraphs describe some of the general functions performed by each of layers **410-420** in BMS **400**.

Enterprise integration layer **410** may be configured to serve clients or local applications with information and services to support a variety of enterprise-level applications. For example, enterprise control applications **426** may be configured to provide subsystem-spanning control to a graphical user interface (GUI) or to any number of enterprise-level business applications (e.g., accounting systems, user identification systems, etc.). Enterprise control applications **426** may also or alternatively be configured to provide configuration GUIs for configuring BMS controller **366**. In yet other embodiments, enterprise control applications **426** can work with layers **410-420** to optimize building performance (e.g., efficiency, energy use, comfort, or safety) based on inputs received at interface **407** and/or BMS interface **409**.

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Building subsystem integration layer **420** may be configured to manage communications between BMS controller **366** and building subsystems **428**. For example, building subsystem integration layer **420** may receive sensor data and input signals from building subsystems **428** and provide output data and control signals to building subsystems **428**. Building subsystem integration layer **420** may also be configured to manage communications between building subsystems **428**. Building subsystem integration layer **420** translate communications (e.g., sensor data, input signals, output signals, etc.) across a plurality of multi-vendor/multi-protocol systems.

Demand response layer **414** may be configured to optimize resource usage (e.g., electricity use, natural gas use, water use, etc.) and/or the monetary cost of such resource usage in response to satisfy the demand of building **10**. The optimization may be based on time-of-use prices, curtailment signals, energy availability, or other data received from utility providers, distributed energy generation systems **424**, from energy storage **427** (e.g., hot TES **242**, cold TES **244**, etc.), or from other sources. Demand response layer **414** may receive inputs from other layers of BMS controller **366** (e.g., building subsystem integration layer **420**, integrated control layer **418**, etc.). The inputs received from other layers may include environmental or sensor inputs such as temperature, carbon dioxide levels, relative humidity levels, air quality sensor outputs, occupancy sensor outputs, room schedules, and the like. The inputs may also include inputs such as electrical use (e.g., expressed in kWh), thermal load measurements, pricing information, projected pricing, smoothed pricing, curtailment signals from utilities, and the like.

According to an exemplary embodiment, demand response layer **414** includes control logic for responding to the data and signals it receives. These responses can include communicating with the control algorithms in integrated control layer **418**, changing control strategies, changing setpoints, or activating/deactivating building equipment or subsystems in a controlled manner. Demand response layer **414** may also include control logic configured to determine when to utilize stored energy. For example, demand response layer **414** may determine to begin using energy from energy storage **427** just prior to the beginning of a peak use hour.

In some embodiments, demand response layer **414** includes a control module configured to actively initiate control actions (e.g., automatically changing setpoints) which minimize energy costs based on one or more inputs representative of or based on demand (e.g., price, a curtailment signal, a demand level, etc.). In some embodiments, demand response layer **414** uses equipment models to determine an optimal set of control actions. The equipment models may include, for example, thermodynamic models describing the inputs, outputs, and/or functions performed by various sets of building equipment. Equipment models may represent collections of building equipment (e.g., sub-plants, chiller arrays, etc.) or individual devices (e.g., individual chillers, heaters, pumps, etc.).

Demand response layer **414** may further include or draw upon one or more demand response policy definitions (e.g., databases, XML, files, etc.). The policy definitions may be edited or adjusted by a user (e.g., via a graphical user interface) so that the control actions initiated in response to demand inputs may be tailored for the user's application, desired comfort level, particular building equipment, or based on other concerns. For example, the demand response policy definitions can specify which equipment may be turned on or off in response to particular demand inputs, how

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long a system or piece of equipment should be turned off, what setpoints can be changed, what the allowable set point adjustment range is, how long to hold a high demand setpoint before returning to a normally scheduled setpoint, how close to approach capacity limits, which equipment modes to utilize, the energy transfer rates (e.g., the maximum rate, an alarm rate, other rate boundary information, etc.) into and out of energy storage devices (e.g., thermal storage tanks, battery banks, etc.), and when to dispatch on-site generation of energy (e.g., via fuel cells, a motor generator set, etc.).

Integrated control layer **418** may be configured to use the data input or output of building subsystem integration layer **420** and/or demand response layer **414** to make control decisions. Due to the subsystem integration provided by building subsystem integration layer **420**, integrated control layer **418** can integrate control activities of the subsystems **428** such that the subsystems **428** behave as a single integrated super-system. In an exemplary embodiment, integrated control layer **418** includes control logic that uses inputs and outputs from a plurality of building subsystems to provide greater comfort and energy savings relative to the comfort and energy savings that separate subsystems could provide alone. For example, integrated control layer **418** may be configured to use an input from a first subsystem to make an energy-saving control decision for a second subsystem. Results of these decisions can be communicated back to building subsystem integration layer **420**.

Integrated control layer **418** is shown to be logically below demand response layer **414**. Integrated control layer **418** may be configured to enhance the effectiveness of demand response layer **414** by enabling building subsystems **428** and their respective control loops to be controlled in coordination with demand response layer **414**. This configuration may advantageously reduce disruptive demand response behavior relative to conventional systems. For example, integrated control layer **418** may be configured to assure that a demand response-driven upward adjustment to the setpoint for chilled water temperature (or another component that directly or indirectly affects temperature) does not result in an increase in fan energy (or other energy used to cool a space) that would result in greater total building energy use than was saved at the chiller.

Integrated control layer **418** may be configured to provide feedback to demand response layer **414** so that demand response layer **414** checks that constraints (e.g., temperature, lighting levels, etc.) are properly maintained even while demanded load shedding is in progress. The constraints may also include setpoint or sensed boundaries relating to safety, equipment operating limits and performance, comfort, fire codes, electrical codes, energy codes, and the like. Integrated control layer **418** is also logically below fault detection and diagnostics layer **416** and automated measurement and validation layer **412**. Integrated control layer **418** may be configured to provide calculated inputs (e.g., aggregations) to these higher levels based on outputs from more than one building subsystem.

Automated measurement and validation (AM&V) layer **412** may be configured to verify that control strategies commanded by integrated control layer **418** or demand response layer **414** are working properly (e.g., using data aggregated by AM&V layer **412**, integrated control layer **418**, building subsystem integration layer **420**, FDD layer **416**, or otherwise). The calculations made by AM&V layer **412** may be based on building system energy models and/or equipment models for individual BMS devices or subsystems. For example, AM&V layer **412** may compare a

model-predicted output with an actual output from building subsystems 428 to determine an accuracy of the model.

Fault detection and diagnostics (FDD) layer 416 may be configured to provide on-going fault detection for building subsystems 428, building subsystem devices (i.e., building equipment), and control algorithms used by demand response layer 414 and integrated control layer 418. FDD layer 416 may receive data inputs from integrated control layer 418, directly from one or more building subsystems or devices, or from another data source. FDD layer 416 may automatically diagnose and respond to detected faults. The responses to detected or diagnosed faults may include providing an alert message to a user, a maintenance scheduling system, or a control algorithm configured to attempt to repair the fault or to work-around the fault.

FDD layer 416 may be configured to output a specific identification of the faulty component or cause of the fault (e.g., loose damper linkage) using detailed subsystem inputs available at building subsystem integration layer 420. In other exemplary embodiments, FDD layer 416 is configured to provide "fault" events to integrated control layer 418 which executes control strategies and policies in response to the received fault events. According to an exemplary embodiment, FDD layer 416 (or a policy executed by an integrated control engine or business rules engine) may shut-down systems or direct control activities around faulty devices or systems to reduce energy waste, extend equipment life, or assure proper control response.

FDD layer 416 may be configured to store or access a variety of different system data stores (or data points for live data). FDD layer 416 may use some content of the data stores to identify faults at the equipment level (e.g., specific chiller, specific AHU, specific terminal unit, etc.) and other content to identify faults at component or subsystem levels. For example, building subsystems 428 may generate temporal (i.e., time-series) data indicating the performance of BMS 400 and the various components thereof. The data generated by building subsystems 428 may include measured or calculated values that exhibit statistical characteristics and provide information about how the corresponding system or process (e.g., a temperature control process, a flow control process, etc.) is performing in terms of error from its setpoint. These processes can be examined by FDD layer 416 to expose when the system begins to degrade in performance and alert a user to repair the fault before it becomes more severe.

Stability Analysis System

Referring now to FIG. 5A, a block diagram of BMS 400 is shown that illustrates network 446 in greater detail, according to some embodiments. FIG. 5A shows, in particular, a stability analysis system 500 communicably coupled to network 446. System 500 may be one of the remote systems and applications 444 discussed above with respect to FIG. 4, for example. The remote systems and applications 444 are also shown as including a dependent analysis system 501 which is communicably coupled to the stability analysis system 500, and can be provided as part of various types of computing resources in various embodiments. In some embodiments, the remote systems and applications 444 are hosted (e.g., implemented) on a computing resources which is a dedicated server, computer, controller, or other processing device. In some embodiments, remote systems and applications 444 are provided on cloud computing resources (e.g., in servers, in distributed cloud architecture, etc.). In some embodiments, systems 500/501 are provide on computing resources locally at the edge (e.g., on-premises with building equipment). For

example, in other embodiments, the remote systems and applications 444 are hosted or implemented by one of client devices 448, or may be a subsystem of BMS controller 366. In this regard, the functionality of systems 500/501, as described herein, may be implemented by any computing resource configured (e.g., programmed) to provide the operations described herein, for example any suitable remote device (e.g., server, etc.) and/or BMS controller 366 (e.g., or other edge device locally at a building), or by a combination of a remote device and BMS controller 366. Accordingly, it will be appreciated by those in the art that the particular layout of the components shown in FIG. 5A-C is one example of the implementation of systems 500/501, and is not intended to be limiting.

BMS controller 366 may be communicably coupled to any number of devices (e.g., building equipment), shown as devices 502-506. Devices 502-506 can include any of the equipment described above with respect to building subsystems 428, including but not limited to electrical equipment, ICT equipment, security equipment, HVAC equipment, lighting equipment, lift/escalators equipment, and fire safety equipment. System 500 may receive operational data from devices 502-506 via BMS controller 366 (e.g., collected by BMS controller 366) and/or network 446. In some embodiments, devices 502-506 may also be communicably coupled directly to network 446. In such embodiments, remote systems such as system 500 may be able to receive operational data directly from device 502-506.

Referring now to FIG. 5B, a block diagram of stability analysis system 500 is shown, according to some embodiments. In general, system 500 functions by processing operational data received from a variety of building equipment (e.g., devices 502-506) to determine a stability index for the building equipment. As briefly discussed above, the stability index is a metric that indicates the stability, and thereby the overall health and efficiency, of building equipment. In this regard, the stability index may be useful in determining whether equipment is running optimally (e.g., efficiently), or whether the equipment is experiencing issues or problems that may lead to excessive wear, inefficiency, failure, etc., if left unaddressed. The stability index may be a useful metric for monitoring system operations and, in some cases, may allow for the early diagnosis and repair of equipment issues.

System 500 includes a computing resource, in particular shown as including a processing circuit 510 that includes a processor 512 and a memory 520. It will be appreciated that these components can be implemented using a variety of different types and quantities of processors and memory. For example, processor 512 can be a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components. Processor 512 can be communicatively coupled to memory 520. While processing circuit 510 is shown as including one processor 512 and one memory 520, it should be understood that, as discussed herein, a processing circuit and/or memory may be implemented using multiple processors and/or memories in various embodiments. All such implementations are contemplated within the scope of the present disclosure.

Memory 520 can include one or more devices (e.g., memory units, memory devices, storage devices, etc.) for storing data and/or computer code for completing and/or facilitating the various processes described in the present disclosure. Memory 520 can include random access memory (RAM), read-only memory (ROM), hard drive storage,

temporary storage, non-volatile memory, flash memory, optical memory, or any other suitable memory for storing software objects and/or computer instructions. Memory 520 can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. Memory 520 can be communicably connected to processor 512 via processing circuit 510 and can include computer code for executing (e.g., by processor 512) one or more processes described herein.

Memory 520 is shown to include a data manager 522. Data manager 522 is generally configured to perform a variety of functions using operational data received from equipment 532 and/or sensors 534 via a communications interface 530, described in detail below. Upon receiving said operational data, data manager 522 can be configured to preprocess and/or store the operational data in a database 528. Subsequently, data manager 522 can retrieve stored data from database 528 for further processing (e.g., by another component of system 500). In this manner, data manager 522 may also be configured to “direct” the flow of data between components of system 500.

Operational data, as discussed herein, can generally include any sort of data or signal regarding the operations of equipment 532. Operational data may include any equipment parameters, measurements, or metrics such as temperature, pressure, status, speed, load, etc. In one example, operational data for a chiller can include fluid outlet and/or inlet temperatures and pressures, compressor speed and energy consumption, compressor oil level and pressure, fan motor energy consumption, etc. In another example, operational data for an AHU could include discharge air temperature, fan motor current, etc. In some embodiments, operational data is received directly from equipment 532 (e.g., from internal or embedded sensors). In other embodiments, operational data is received from sensors 534 or from any combination of equipment 532 and sensors 534.

Equipment 532 may include any number of devices (i.e., equipment) associated with any of building subsystems 428, as described above, such as HVAC equipment (e.g., ducts, VAVs, AHUs, etc.), security equipment (e.g., cameras, card readers, etc.), fire safety equipment (e.g., alarms, sprinklers, etc.), etc. Likewise, sensors 534 may include any number and type of suitable sensors for measuring operational parameters associated with the equipment of BMS 400. For example, sensors 534 can include temperature sensors, pressure sensors, smoke detectors, access control readers, security cameras, etc. In various embodiments, sensors 534 may be coupled to, embedded in, or collocated with equipment 532. In other words, sensors 534 may be stand-alone (i.e., external or remote) sensors, or may be embedded within equipment 532.

In some embodiments, data manager 522 preprocesses operational data received from equipment 532 and/or sensors 534 prior to storage (e.g., in database 528) and/or prior to further manipulation by other components of system 500. In such embodiments, preprocessing can include formatting, normalizing, or otherwise modifying or altering the data such that the data is in an appropriate format for storage and/or further manipulation. In one example, data manager 522 receives data in a first format or unit of measure and converts the data to a second format or unit of measure. In some embodiments, preprocessing may include time stamping the data based on a time of receipt, before the data is stored in database 528.

In some embodiments, data manager 522 stores data that is not formatted, normalized, or otherwise modified (i.e., raw data) in database 528. In such embodiments, data manager 522 may retrieve the data at later time (e.g., prior to routing the data to another component of memory 520) for preprocessing. In such embodiments, data manager 522 may format, normalize, or otherwise modify the data upon retrieval of the data from database 528. For example, raw operational data stored in database 528 may be retrieved at a regular time interval for further analysis, in response to a user request for a particular report, to generate or update a user interface, etc. In some cases, preprocess the retrieved data may include backfilling or forward-filling datasets, as discussed in further detail below with respect to step 602 of process 600.

It will be appreciated that, as described herein, database 528 can be implemented as a single database or multiple separate databases working together. Database 528 can be configured to maintain a wide variety of data and data types associated with system 500. For example, database 528 can generally maintain raw and/or preprocessed equipment operating data, along with other historical data relating to the operations of equipment 532. It will also be appreciated that, while shown as an internal component of system 500, database 528 could also be implemented at least partially by one or more external databases (e.g., hosted by one or more remote devices).

Still referring to FIG. 5B, memory 520 is also shown to include an equipment stability analyzer 524. In general, equipment stability analyzer 524 may receive operational data (e.g., raw or preprocessed) from data manager 522 and, in some cases, may retrieve stored data directly from database 528. Equipment stability analyzer 524 is generally configured to perform various analyses, using current or historical operational data, in order to determine a stability index for a building device (e.g., equipment 532). As described above, the stability index may provide an indication of the health and/or efficiency of a building device, allowing for improved and preemptive maintenance of failing or problematic equipment, thereby leading to decreased downtime and repair costs.

As described below in greater detail with respect to FIG. 6, equipment stability analyzer 524 may apply the operational data to a stability index model. In some embodiments, the stability index model includes a series of algorithms used to calculate a stability index value from the operational data. In some embodiments, equipment stability analyzer 524 utilizes machine learning algorithms or neural networks to train and improve the stability index calculation. Based on the stability index, equipment stability analyzer 524 may determine one or more automated (i.e., automatic) responses. In some cases, equipment stability analyzer 524 may cause a user interface (UI) generator 526 to generate a graphical user interface based on the stability index. In other cases, equipment stability analyzer 524 may determine and transmit (e.g., via data manager 522 and/or communications interface 530) one or more control decisions to equipment 532, causing one or more devices of equipment 532 to cease operations, adjust operating parameters, etc. In yet other cases (e.g., where the stability index is below a threshold), equipment stability analyzer 524 may automatically determine and/or schedule an appropriate maintenance or service request to correct problems with equipment stability analyzer 524.

UI generator 526 may be configured to dynamically generate graphical user interfaces for presenting a wide variety of information regarding system 500. As an example,

UI generator **526** may generate a graphical user interface to present the stability index value(s) calculated by equipment stability analyzer **524**. In another example, UI generator **526** can generate a graphical user interface that includes a “health dashboard,” presenting an overview of the stability and/or health of a plurality of equipment or devices. UI generator **526** may be implemented as a webserver that can store, process, and deliver web pages (e.g., HTML documents) to a web browser of a user device **546**, or as an application on a user device **546** (e.g., desktop application, mobile application), for example, although it will be appreciated that UI generator **526** can be implemented in any other suitable manner as well. UI generator **526** may generally receive inputs (e.g., HTTP requests) from one or more of client devices **448**, thereby providing an interface for a user to interact with system **500**.

Still referring to FIG. **5B**, system **500** is also shown to include communications interface **530** configured to exchange data with a plurality of remote systems and/or devices. In particular, communications interface **530** may allow system **500** to exchange data via network **446**, as previously described. In this manner, system **500** may transmit and/or receive data from any of the components of BMS **400** via communications interface **530**. In some embodiments, communications interface **530** may also facilitate the conversion and/or translation of various data or communication types or formats, to allow system **500** to interface with a wide variety of external systems and devices.

In some embodiments, system **500** may communicate directly with one or more client devices **448** via communications interface **530**. In other embodiments, system **500** may communicate with client devices **448** via network **446**. As described above, client devices **448** may include any electronic device that allows a user to interact with system **500** and/or BMS **400** through a user interface. Examples of user devices include, but are not limited to, mobile phones, electronic tablets, laptops, desktop computers, workstations, and other types of electronic devices. One or more of client devices **448** may present (i.e., display) a plurality of user interfaces generated by system **500** including, in some cases, an interface that presents the stability index for one or more devices.

In some embodiments, system **500** may also exchange data directly with equipment **532** and/or sensors **534** via communications interface **530**. In some such embodiments, communications interface **530** may include a wired and/or wireless BACnet interface in addition to other types of communications interfaces (e.g., Modbus, LonWorks, DeviceNet, XML, etc.) to facilitate communication with equipment **532** and/or sensors **534**. In other embodiments, system **500** may exchange data with equipment **532** and sensors **534** via network **446**, as described above with respect to FIG. **5A**. In such embodiments, data from equipment **532** and/or sensors **534** may be received and otherwise manipulated by BMS controller **366**, before being transmitted to system **500** via network **446**. Accordingly, it will be appreciated that equipment **532**, sensors **534**, and/or client devices **448** may be communicably coupled to system **500** in any suitable manner, and that the particular configuration shown in FIG. **5B** is not intended to be limiting.

Referring now to FIG. **5C**, a dependent analysis system **501** is shown, according to some embodiments. As illustrated in FIG. **5A**, the dependent analysis system **501** may be included with remote systems and applications **444**. FIGS. **5B-C** also show that the dependent analysis system **501** is communicable with the stability analysis system **500**. The

dependent analysis system **501** performs operations dependent on values of the stability index output by the stability analysis system **500**.

As illustrated in FIG. **5C**, the dependent analysis system **501** provides a computing resource, shown as including a processing circuit **511**, which includes processor **513** and memory **521**. In some embodiments, the stability analysis system **500** and the dependent analysis system **501** are implemented using a unified processing circuit programmed to execute the operations attributed to both systems **500/501** herein. In some embodiments, the processor **513** includes one or more processors and the memory **521** includes one or more computer-readable media storing program instructions that, when executed, cause the processor **513** to perform the operations attributed herein to the dependent analysis system **501**.

The memory **521** is shown as including a data manager **523**, a performance analyzer **525**, a user interface generator **527**, and a database **529**. The dependent analysis system **501** is also shown as including communications interface **531** which is communicable with one or more client devices **448**, network **446**, equipment **532**, sensors **534**, and stability analysis system **500**.

The data manager **523** preprocesses operational data received from equipment **532** and/or sensors **534** prior to storage (e.g., in database **529**) and/or prior to further manipulation by other components of system **500**. In such embodiments, preprocessing can include formatting, normalizing, or otherwise modifying or altering the data such that the data is in an appropriate format for storage and/or further manipulation. The database **529** may be configured to maintain a wide variety of data and data types associated with system **501**. For example, database **529** can generally maintain raw and/or preprocessed equipment operating data, along with other historical data relating to the operations of equipment **532**. It will also be appreciated that, while shown as an internal component of system **500**, database **529** could also be implemented at least partially by one or more external databases (e.g., hosted by one or more remote devices).

The performance analyzer **525** is configured to perform one or more of various performance analyses using or in response to a characteristics of stability index values from the stability index system **500**. The performance analyzer can use data from the database **529** and/or the data manager **523** to execute various algorithms, simulations, artificial intelligence algorithms, etc. to provide insights into equipment performance, energy usage, equipment faults, etc., various examples of which are provided in detail below with reference to FIGS. **10-12**.

As one example, the performance analyzer **525** may be configured similar to the time varying performance indication system of U.S. Pat. No. 11,092,954, the entire disclosure of which is incorporated by reference herein. For example, the performance analyzer **525** may obtain data points of one or more monitored variables measured by the equipment **532** and a plurality of status codes generated by the equipment **532**. The performance analyzer **525** may also obtain a plurality of equipment specific parameters that are specific to the equipment **532** (e.g., to individual units or devices of equipment **532**). In such examples, the performance analyzer **525** may perform a plurality of first performance checks for the connected equipment using the plurality of status codes from the past *N* time units, and also perform a plurality of second performance checks for the connected equipment using the data points of the plurality of monitored variables, the status codes, and the plurality of

connected equipment specific parameters. The performance analyzer 525 may then generate a plurality of individual performance check indicators based on the first and second performance checks using a plurality of first weights determined based on different timings, and determine an overall performance index for the equipment 532 using the plurality of individual performance check indicators and a plurality of second weights.

The user interface generator 527 is configured to generate a user interface reporting outputs of the performance analyzer 525 and various other data relating to the system 501. The user interface generator 527 may be implemented as a webserver that can store, process, and deliver web pages (e.g., HTML documents) to a web browser of a user device 546, or as an application on a user device 546 (e.g., desktop application, mobile application), for example. User interface generator 527 UI may generally receive inputs (e.g., HTTP requests) from one or more of client devices 448, thereby providing an interface for a user to interact with system 501.

In some embodiments, results of operations of the performance analyzer 525 are control outputs, control signals, setpoints, etc. for the equipment 532. In such embodiments, such results are transmitted to the equipment 532 (e.g., to controllers thereof) via the communications interface 531, thereby affecting operation of the equipment 532.

Referring now to FIG. 6, an example process 600 for determining a stability index for building equipment (e.g., equipment 532) is shown, according to some embodiments. In some embodiments, process 600 is implemented by system 500. It will be appreciated that certain steps of process 600 may be optional and, in some embodiments, process 600 may be implemented using less than all of the steps and/or different steps than are shown in FIG. 6. As described below, process 600 may include steps for determining a stability index for a building device (e.g., a chiller, a cooling tower, a boiler, an AHU, etc.) that provides a heated or cooled fluid (e.g., water, coolant, air, etc.), but it will be appreciated that process 600 may also be used to calculate a stability index for other building equipment.

In one example, where process 600 is implemented to determine a stability index for a chiller (e.g., chiller 102), the stability index may indicate how efficiently the chiller is operating, and thereby how stable the chiller is. A chiller's primary goal is to provide chilled water for building cooling. To meet the chilled water requirements, a chiller must deliver chilled water that is at a temperature typically between 38 to 46° F. While the dynamics of chiller operations can be complex, and chillers can be affected by many different issue, most problems will affect the chillers ability to maintain chilled water setpoint and, accordingly, may cause a loss of chilled water stability. The stability index described herein may provide an indicator of these chiller instabilities. In other embodiments, the stability index may likewise indicate an efficiency of other building equipment such as a boiler, cooling tower, AHU, etc.

At step 602, time series data regarding at least one operational parameter of a building device or equipment (i.e., equipment 532) is received. Time series data, as described herein, is generally a data set that includes a series of data points indexed in time order. In some embodiments, operational data (e.g., data regarding at least one operational parameter of a device) is collected in a series of discrete, equally-spaced points in time (i.e., time steps). In other words, the operational data for building equipment, as described above with respect to FIG. 5B, is typically discrete-time data that is collected and/or recorded at a series of time periods.

In some embodiments, such time series operational data is received directly from building equipment (e.g., from internal or embedded sensors of equipment 532). In other embodiments, operational data is received from one or more external or remote sensors, or from a combination of internal and external sensors. In some such embodiments, where operational data is received from building equipment and/or sensors, the data may be time stamped with a time of receipt. In some cases, time stamped operational data is stored in a database (e.g., database 528) for later retrieval and manipulation. Accordingly, in some embodiments, the time series operational data is retrieved from a database (e.g., database 528), having previously been received and stored (e.g., by data manager 522).

In some embodiments, the first operational parameter is an outlet fluid temperature for building equipment that provides heated or cooled fluid (e.g., water, air, coolant, oil, etc.). In particular, the first operational parameter may be a chilled water supply temperature for a chiller, cooling tower, or other device, a heated water supply temperature for a boiler or other similar device, or even a discharge air temperature for an AHU or rooftop unit. Accordingly, the time series data received at step 602 may include fluid supply/discharge temperatures recorded at an outlet of the chiller, cooling tower, boiler, AHU, etc. The time series data may also include a setpoint for the outlet fluid temperature. In the example of a chiller, the time series data may include both a chilled water supply temperature and chilled water supply temperature setpoint.

At step 604, a time delta is calculated for one or more time steps of the time series data. A time delta may indicate the length of an individual time step, based on a difference between a first and a second time step. In some embodiments, the time steps of the time series data are represented as a series of t_i from $t_1 \dots t_n$, where n is the total number of samples in the data set. Generally, the time delta for an individual time step is calculated as:

$$TD_i = t_i - t_{i-1}$$

where TD_i is the time delta at step i , and t_i and t_{i-1} are a first and second time step. In this case, t_{i-1} represents the time step immediately before t_i , and the difference between these times steps is the time delta. The calculation of a time delta for each time step is illustrated in FIG. 7, which is described in detail below.

In some embodiments, the time delta is calculated only after the operational data has been preprocessed, although it will be appreciated that preprocessing is not always required. Preprocessing can include formatting, normalizing, or otherwise modifying or altering the data such that the data is in an appropriate format for storage and/or further manipulation. In various embodiments, operational data is received in a first format or unit of measure and converted to a second format or unit of measure. In some embodiments, preprocessing includes forward- or back-filling a data set. In some cases, for example, data may only be recorded when an operational parameter changes, or it may be desirable to limit the number of data points of a time series data set. In such example cases, certain data points may be removed or added to the time series data during preprocessing.

At step 606, an absolute deviation (error) between a setpoint and a recorded value is calculated for the one or more time steps. The absolute deviation may indicate how closely a recorded value (i.e., measurement) for an operational parameter is tracking the desired setpoint. The absolute deviation is calculated taking the absolute value of the

difference between a measured or recorded value for a first operational parameter and a setpoint for the first operational parameter. Mathematically, this is represented as:

$$ABD_i = |FST_i - FST_{sp_i}|$$

where ABD_i represents the absolute deviation at step i , FST_i represents the measured or recorded value (i.e., data point) for the first operational parameter (e.g., fluid supply temperature) at step i , and FST_{sp_i} represents the setpoint for the first operational parameter at step i . Hence, the calculated absolute values form a series $ABD_1 \dots ABD_n$ for a data set that includes a plurality of time steps. The calculation of an absolute deviation for each time step is illustrated in FIG. 7, which is described in detail below.

To continue the previous example regarding a chiller, the absolute value (ABD_i) at a time step may indicate how close the chilled water supply temperature is tracking to a setpoint. While a certain amount of fluctuation between the measured temperature and the setpoint is to be expected, large discrepancies between the measured temperature and setpoint can indicate inefficiencies with the chiller, particularly if the chiller fails to track the setpoint over a long period of time. In some cases, deviations between a fluid supply temperature and a setpoint can even indicate a chiller that is experiencing a malfunction or a failure, or that the chiller may fail in the near future.

At step 608, a trapezoidal area is calculated for one or more time steps. Generally, the trapezoidal areas are calculated based on the time deltas calculated for the one or more time steps at step 604 and the absolute deviations calculated at step 606. The trapezoidal area may indicate how closely the setpoint was tracked over the time step, where smaller trapezoidal areas indicate that the measured or recorded operational data was more closely tracking the setpoint. A trapezoidal area is calculated as:

$$Area_i = (ABD_i + ABD_{i+1}) \times \left(\frac{TD_i}{2} \right)$$

where $Area_i$ is the trapezoidal area for a time step i , ABD_i is the absolute deviation at time step i and ABD_{i+1} is the absolute deviation calculated at the time step immediately following time step i . The calculation of a trapezoidal area is illustrated in FIG. 7, which is described in detail below.

At step 610, the calculated trapezoidal areas are filtered to exclude time steps where a second operational parameter is below a threshold. In some embodiments, the areas are filtered by setting the absolute deviation (ABD_i) to zero for time steps where the second operational parameter is below a threshold. In other embodiments, trapezoidal areas corresponding to a time step where the second operational parameter is below a threshold are simply not included in the summation at step 612, described below.

In some embodiments (e.g., for a chiller), the second operational parameter is a current for a motor (e.g., the compressor motor) of the chiller, typically presented as a percentage of full load amperage. In one example, if the full load amperage is below a threshold at a particular time step, such as 5% of maximum full load amperage, then it may be assumed that the compressor, and thereby the chiller, was not turned on (i.e., operating). Accordingly, any data recorded during this time step may be ignored.

At step 612, the trapezoidal areas are summed for a subset of the one or more time steps over a time horizon. The time horizon may be any suitable amount of time (e.g., a day, a

month, an hour, etc.). The summation of the trapezoidal areas may be represented as a unit area (UAREA). The UAREA is calculated as:

$$Uarea_j = \sum_{i \in U_j} Area_i$$

where U_j is unit time j . It will be appreciated that j can be any suitable amount of time, such as minutes, hours, days, etc., and may include a plurality of smaller time steps. In one example, j is an hour and includes a plurality of time steps each corresponding to a one minute interval.

At step 614, a stability index is calculated for the building equipment based on the summation of trapezoidal areas (e.g., UAREA) calculated at step 612. The stability index for unit time j is calculated as:

$$SI_j = 100 - (\lambda \times Uarea_j)$$

where λ is an optional scale factor. In some embodiments, the scale factor is adjustable and can be tuned (e.g., based on the type of building equipment). The scale factor acts as a penalty for the UAREA value calculated at step 612. In this regard, a larger scale factor will more heavily penalize the UAREA. For example, calculating the stability index for a chiller using a scale factor of 10 will cause a penalty of 10 to the stability index for an average deviation of 1° F. from a setpoint. In some embodiments, a scale factor of 30 is used for a chiller.

At step 616, an automatic response is initiated based on the stability index. In some embodiments, the automatic response includes the generation of a graphical user interface. The graphical user interface may be presented via one or more client devices 448, and may include a graphical or textual representation of the stability index, as shown in FIG. 8 below. Additionally, the graphical user interface may include other visual elements that present information relating to the first or second operational parameters, as described above. In some embodiments, the graphical user interface may include a “health dashboard” that provides an overview of the health or stability of multiple building devices in a single interface.

In some embodiments, such as when the stability index is below a threshold, the automatic response process may include generating a work order or a ticket, and/or automatically scheduling maintenance or service for a building device. In such embodiments, the threshold may be previously defined by a user (e.g., a system administrator). In some embodiments, building equipment with a stability index below a threshold may be flagged, such that a user may quickly identify the equipment and may determine additional action (e.g., scheduling maintenance). In some embodiments, the automatic response also includes generating a message or alert, and transmitting the message or alert to a remote device (e.g., client devices 448). For example, system 500 may automatically generate a text message or email, and may transmit the text message or email to a user device associated with a maintenance technician, system administrator, etc.

Referring now to FIG. 7, an example graph of a plurality of trapezoidal areas over a plurality of time periods is shown, according to some embodiments. As mentioned above, FIG. 7 may provide a visual representation of various steps of process 600. As shown, for example, the graph includes six time periods, with five of the six periods having a trapezoidal area. A first time period on the far left of FIG. 7 illustrates various variables described above with respect to FIG. 6. For example, the trapezoidal area for this first time period is shown to be calculated from an absolute deviation ABD_i at a first time step t_i and absolute deviation ABD_{i+1} at

a second time step t_{i+1} . Here, the time delta TD_i is shown as the different between time t_i and t_{i+1} . Also shown in FIG. 7 is a particular time period where a trapezoidal area is not calculated. As discussed above, this particular time period may have been filtered between a second operational parameter fell outside of a threshold.

Referring now to FIG. 8, an example interface **800** for presenting a stability index and various other operational data associated with building equipment is shown, according to some embodiments, interface **800** is an interface generated by UI generator **526** and presented via one or more client devices **448**, as described above. In general, interface **800** presents the stability index and any additional operational data via a trend line, or other suitable graphic, over time. Accordingly, interface **800** provides, in a single interface, an overview and/or history of device performance and stability.

It will be appreciated that the elements of interface **800** may include any suitable graphical or textual elements for presenting operating data over time. However, in the example shown, interface **800** includes a plurality of graphs **802-806**, which include trend lines corresponding to various operating parameters of a device. Graph **802**, for example, includes a trend line corresponding to a calculated hourly area for a device. In other words, graph **802** indicates the area from the summation of a plurality of trapezoidal areas, as previous described, over a one-hour period. Graphs **804** and **806** include trend lines for a stability index value and full load amps for the device, also previous described. In this case, the x-axis of each of graphs **802-806** is a number of hours (i.e., time). The number of hours may indicate, for example, a total number of operating hours, or a number of hours since a first data point was recorded.

In a non-limiting example, interface **800** may present trend data for a chiller (e.g., chiller **102**), as described above. In this example, graph **802** would include a trend line corresponding to the summation of trapezoidal areas calculated at each of a plurality of time intervals based on a deviation between a setpoint and a recorded value for an outlet water temperature of the chiller. Graph **804** may present a trend line corresponding to a stability index for the chiller, also calculated at each of a plurality of time intervals. As shown, the stability index value trend line is generally inversely proportional to the area trend line of graph **802**. Graph **806** may include a trend line corresponding to the measured full load amperage (FLA) of a pump or compressor associated with the chiller. In this case, the FLA appears to stay just below 50% of a maximum value.

Referring now to FIG. 9, an example interface **900** for presenting chilled water temperatures over time is shown, according to some embodiments. Interface **900** may be an interface generated by UI generator **526** and presented via one or more client devices **448**, for example. Like interface **800** described above, interface **900** also presents operational data for a device or equipment over time. In this case, interface **900** provides a graphical representation of recorded values and setpoints for a first operational parameter of a device, thereby indicating how closely the device is following the setpoints.

In the example shown, interface **900** provides a graphical representation of chilled water temperature measurements ("CHWST") and chilled water temperature setpoints ("CHWST SP") over a time period of eight days, from August 28th to September 6th. Chilled water temperature measurements, in this case, are represented by a solid black line while chilled water temperature setpoints are represented by a white line. Here, the chilled water temperature setpoint

was set at roughly 43° F., with a slight deviation on September 2nd. The recorded chiller water temperatures are shown to fluctuate around the setpoint value, typically by about $\pm 3^\circ$ F. At certain time periods on September 2nd, 3rd, and 4th, the recorded chilled water temperature is shown to fluctuate as much as $+14^\circ$ F.

Stability Index Processes

Referring now to FIG. 10, a flowchart of a process **1000** for using a stability index is shown, according to some embodiments. Process **1000** can be executed by system **400** of FIG. 5A, for example by remote systems and applications **444**, according to various embodiments. In some examples, process **1000** is executed by a combination of (collaboration between) stability analysis system **500** and dependent analysis system **501**.

At step **1002**, a current value of the stability index is calculated (the current stability index). The current value of the stability index can be calculated as described above with reference to FIGS. 6-9, for example by the stability analysis system **500**. The current value of the stability index of step **1002** is calculated through a current time step based on data for a set of preceding time steps as described in detail above.

At step **1004**, the current stability index is compared to a criterion. Step **1004** can be executed by the dependent analysis system **501**, for example. Various criteria can be used in various embodiments. For example, in some embodiments the criterion is a threshold value, such that the current stability index satisfies the criterion if the current stability index is greater than the threshold value and does not satisfy the criterion if the current stability index is less than the threshold value (or vice versa) (e.g., greater than a threshold stability, less than a threshold stability). As another example, the criterion may be based on a statistical metric of past stability index values, for example an average stability index, a standard deviation from the average stability index, etc.

At step **1006**, an analysis of equipment performance is executed in response to the current stability index satisfying the criterion. If the current stability index does not satisfy the criterion, the analysis of equipment performance is skipped, omitted, prevented, etc. Executing the analysis of equipment performance can include collecting and processing data for the current time step (or up to the current time step) and applying an algorithm, model, artificial intelligence program, etc. to such data.

In some examples, step **1006** includes executing an equipment health check. The equipment health check can include calculating a performance index as described in U.S. Pat. No. 11,092,954, filed Jan. 10, 2019, the entire disclosure of which is incorporated by reference herein. Conditioning execution of equipment health checks on the stability index satisfying a criterion can avoid misleading results that may otherwise be caused by executing health checks during start-up or transitory periods. In some embodiments, executing an equipment health check includes running a control routine to influence operation of the equipment to facilitate assessment of equipment health.

In some examples, the equipment includes a chiller. The stability index can be assessed with respect to leaving chilled water (LCHW) temperature setpoints. In such embodiments, step **1006** can include comparing variable speed drive (VSD) speed and lift (e.g., the difference between condenser refrigerant pressure and evaporator refrigerant pressure) to identify opportunities to improve energy utilization (e.g., via condenser relief) and/or to detect or diagnose faults or inefficiencies in condenser performance.

In some examples, the stability index is assessed with respect to a condition of condenser water and can indicate suspect cooling tower performance. In such examples, step 1006 includes assessing cooling tower performance, flow, or other conditions.

In some examples, step 1006 includes determining that an optimization or other control algorithm is commanding setpoints to change at too high of a rate of change (thereby causing undesirable amounts of instability). For example, if the stability index does not satisfy a criterion for a time that includes greater than a threshold frequency of setpoint changes, step 1006 may diagnose that a control algorithm is generating setpoint changes which are too frequent to allow the equipment to control to such setpoints. Step 1006 can then include tuning parameters, runtimes, constraints, etc. of an optimization program or other control algorithm to reduce the frequency of setpoint changes to a level that achieves less than a threshold level of instability (as indicated by the stability index).

In some examples, step 1006 includes predicting equipment faults or failure using one or more of various data. Performing such predictions in stable states (e.g., when the stability index satisfies a criterion in step 1004) can help avoid false alarms and incorrect fault detections that may otherwise result from assessing data corresponding to unstable operations. Prediction indicators may include refrigerant level, VO, IG, HGBP, VGD, speed, voltage, current, a ratio of change in pressure to pressure, lift, distortion, and subcooler/flash gas drop leg to ECW.

At step 1008, a result of the analysis is output and, in some embodiments, equipment is controlled based on a result of the analysis. The analysis may be output via a user interface, for example a dashboard, visualization, etc. provided by the user interface generator 527. In some embodiments, a result of the analysis provides an on/off decision, setpoint, or other control decision for equipment. In some embodiments, the result of the analysis is a retrained system model, and step 1008 includes causing the retrained system model to be used in model predictive control of equipment. Various tangible, practical outcomes of step 1008 are possible in various embodiments.

Referring now to FIG. 11, a process 1100 involving a time series of stability indices is shown, according to some embodiments. Process 1100 can be executed by system 400 of FIG. 5A, for example by remote systems and applications 444, according to various embodiments. In some examples, process 1100 is executed by a combination of (collaboration between) stability analysis system 500 and dependent analysis system 501.

At step 1102, a time series of stability indices is collected by repeatedly calculating the current stability index over time. At each time step, a current value of the stability index can be calculated (i.e., based on data through that time step), for example as described in detail above. That value can then be stored with a time stamp indicating the corresponding time step, for example. A time series of stability index values (stability indices) can thereby be collected (e.g., by the stability analysis system 500).

At step 1104, the time series of stability indices is compared to a criterion (or to multiple criteria). Various criteria are possible in various embodiments. As one example, a sum of a certain number of stability indices (e.g., the last 10, 20, 50, 100, etc. values) can be compared to a threshold value such that the criterion is satisfied if the sum is greater than the threshold value (or less than the threshold value in other examples). Another example, a criterion may be satisfied upon occurrence of a certain number of

stability index values greater than a threshold (e.g., by increasing a counter value each time the current stability index value is greater than the threshold). As another example, an average of a number of stability indices can be compared to a threshold value. As another example, step 1104 can include determining whether the values of the stability indices are trending upwards or downwards over time, for examples such that satisfying the criterion requires upward trending values (or downward trending values in other examples). Various statistical analyses or other logic can be applied to the time series of stability indices in step 1104 in various embodiments.

At step 1106, an analysis of equipment performance is executed in response to the time series of stability indices satisfying the criterion. If the time series of stability indices do not satisfy the criterion, the analysis of equipment performance (step 1106) is omitted, skipped, prevented, etc. Executing the analysis of equipment performance can include collecting and processing data for the current time step (or up to the current time step) and applying an algorithm, model, artificial intelligence program, etc. to such data.

In some examples, step 1106 includes executing an equipment health check. The equipment health check can include calculating a performance index as described in U.S. Pat. No. 11,092,954, filed Jan. 10, 2019, the entire disclosure of which is incorporated by reference herein. Conditioning execution of equipment health checks on the stability index satisfying a criterion can avoid misleading results that may otherwise be caused by executing health checks during start-up or transitory periods. In some embodiments, executing an equipment health check includes running a control routine to influence operation of the equipment to facilitate assessment of equipment health.

In some examples, the equipment includes a chiller. The stability index can be assessed with respect to leaving chilled water (LCHW) temperature setpoints. In such embodiments, step 1106 can include comparing variable speed drive (VSD) speed and lift (e.g., the difference between condenser refrigerant pressure and evaporator refrigerant pressure) to identify opportunities to improve energy utilization (e.g., via condenser relief) and/or to detect or diagnose faults or inefficiencies in condenser performance.

In some examples, the stability index is assessed with respect to a condition of condenser water and can indicate suspect cooling tower performance. In such examples, step 1006 includes assessing cooling tower performance, flow, or other conditions.

In some examples, step 1106 includes determining that an optimization or other control algorithm is commanding setpoints to change at too high of a rate of change (thereby causing undesirable amounts of instability). For example, if the stability indices do not satisfy a criterion through a time period that includes greater than a threshold frequency of setpoint changes, step 1006 may diagnose that a control algorithm is generating setpoint changes which are too frequent to allow the equipment to control to such setpoints. Step 1106 can then include tuning parameters, runtimes, constraints, etc. of an optimization program or other control algorithm to reduce the frequency of setpoint changes to a level that achieves less than a threshold level of instability (as indicated by the stability index).

In some examples, step 1106 includes predicting equipment faults or failure using one or more of various data. Performing such predictions in stable states (e.g., when stability indices satisfy criteria in step 1104) can help avoid

false alarms and incorrect fault detections that may otherwise result from assessing data corresponding to unstable operations. Prediction indicators may include refrigerant level, VO, IGV, HGBP, VGD, speed, voltage, current, a ratio of change in pressure to pressure, lift, distortion, and sub-cooler/flash gas drop leg to ECW.

At step **1108**, result of the analysis is output and, in some embodiments, equipment is controlled based on a result of the analysis. The analysis may be output via a user interface, for example a dashboard, visualization, etc. provided by the user interface generator **527**. In some embodiments, a result of the analysis provides an on/off decision, setpoint, or other control decision for equipment. In some embodiments, the result of the analysis is a retrained system model, and step **1008** includes causing the retrained system model to be used in model predictive control of equipment. Various tangible, practical outcomes of step **1108** are possible in various embodiments.

Referring now to FIG. **12**, a flowchart of a process **1200** involving a time series of stability indices is shown, according to some embodiments. Process **1200** can be executed by system **400** of FIG. **5A**, for example by remote systems and applications **444**, according to various embodiments. In some examples, process **1200** is executed by a combination of (collaboration between) stability analysis system **500** and dependent analysis system **501**.

At step **1202**, a time series of stability indices is collected by repeatedly calculating the current stability index over time. At each time step, a current value of the stability index can be calculated (i.e., based on data through that time step), for example as described in detail above. That value can then be stored with a time stamp indicating the corresponding time step, for example. A time series of stability index values (stability indices) can thereby be collected (e.g., by the stability analysis system **500**).

At step **1204**, the stability indices in the time series are compared to criteria. Each value of the stability index (i.e., associated with a given time step) can be compared to a criterion (e.g., compared to a threshold value), such that step **1204** includes generating a time series of results of such a comparison. For example, step **1204** may result in a time series of binary values (e.g., 1 or 0) indicating whether the criterion is satisfied for the corresponding time step (e.g., 1) or not satisfied for the corresponding time step (e.g., 0).

At step **1206**, operating data is collected. The operating data can be raw or preprocessed operating data, for example relating to operation of equipment (setpoints, power consumption, measured conditions, etc.), building conditions (e.g., temperature, humidity, pressure, airflow, light level, noise level, etc.), energy data (e.g., energy used, carbon emissions attributable to building operations, pollution attributable to energy usage), or other data (e.g., occupancy data, building schedules, weather data, etc.). In the example of FIG. **12**, the operating data is time series data, such that values of any variables represented in the operating data are identified as corresponding to particular time steps. Collecting operating data in step **1206** can include receiving data at the data manager **523** of dependent analysis system **501** via communications interface **531** of FIG. **5C**, managing, organizing, etc. such operating data by the data manager **523**, and storage of such operating data in database **529** of dependent analysis system **501**, in example embodiments.

At step **1208**, the operating data is filtered based on the result of comparing the stability indices in the time series of stability indices to criteria at step **1204**. For example, operating data can be removed, ignored, deleted, etc. for time steps when the stability index did not satisfy the

criterion, while operating data is preserved, marked for use, passed through the filter, etc. for steps when the stability index satisfied the criterion. In such examples, filtered operating data resulting from step **1208** includes operating data for the time steps when the stability indices satisfied the criteria and does not include operating data for the time steps when the stability indices did not satisfy the criteria. In other examples, step **1208** includes weighting operating data or otherwise processing the operating data to deemphasize, reduce the effect of, etc. operating data for time steps when the stability indices did not satisfy the criteria.

At step **1210**, the filtered operating data is used to affect operations, for example used to control building equipment. In some embodiments, the filtered operating data is used to train a system model or adjust a model of system operation which is used in a model predictive control approach to generate setpoints for building equipment. In other examples, the filtered operating data is used to generate one or more metrics, key performance indicators, alerts, alarms, maintenance or supply orders, etc., for example as in the examples described above with reference to steps **1006** and **1106** as described above with reference to FIGS. **10-11**. In one example, step **1210** includes calculating a performance index for connected equipment (as in U.S. Pat. No. 11,092,954, filed Jan. 10, 2019, the entire disclosure of which is incorporated by reference herein) using the filtered operating data. Process **1200** thereby uses stability indices and criteria relating thereto to select a suitable set of operating data for performing further analytics or operations, which can increase the accuracy, reliability, trustworthiness, repeatability, etc. of such analytics or operations. For example, process **1200** can be used to suppress alarms (e.g., fault suppression) during periods in which the system is unstable (as indicated by stability indices not satisfying criteria).

35 Configuration of Exemplary Embodiments

The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products including machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise

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RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

What is claimed is:

1. A method for monitoring and controlling building equipment, comprising:

obtaining an actual value of a condition affected by operating the building equipment;
operating the building equipment using a setpoint for the condition;
calculating a stability index based on a timeseries of errors between the actual value and the setpoint;
comparing the stability index to a criterion; and
in response to the stability index satisfying the criterion, modifying an operation of the building equipment and operating the building equipment in accordance with the modified operation.

2. The method of claim 1, wherein the condition comprises at least one of temperature, humidity, pressure, flow rate, light level, or noise level.

3. The method of claim 1, wherein calculating the stability index comprises calculating a sum of a plurality of trapezoidal areas having sides corresponding to the timeseries of errors.

4. The method of claim 1, wherein comprising performing a fault detection or diagnostic procedure for the building equipment in response to the stability index satisfying the criterion.

5. The method of claim 1, wherein modifying the operation comprises modifying a predictive control strategy for the building equipment and wherein operating the building equipment in accordance with the modified operation comprises operating the building equipment in accordance with the modified predictive control strategy.

6. The method of claim 1, wherein the building equipment comprises a chiller and the method comprises calculating a chiller performance index.

7. The method of claim 1, wherein comparing the stability index to the criterion comprises determining whether the stability index is greater than a predefined threshold.

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8. One or more non-transitory computer-readable media storing program instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising:

obtaining an actual value of a condition affected by operating building equipment;
operating the building equipment using a setpoint for the condition;
calculating a stability index based on a timeseries of errors between the actual value and the setpoint;
comparing the stability index to a criterion; and
in response to the stability index satisfying the criterion, modifying an operation of the building equipment and operating the building equipment in accordance with the modified operation.

9. The one or more non-transitory computer-readable media of claim 8, wherein the condition comprises at least one of temperature, humidity, pressure, or flowrate.

10. The one or more non-transitory computer-readable media of claim 8, wherein calculating the stability index comprises calculating a sum of a plurality of trapezoidal areas defined by the timeseries of errors.

11. The one or more non-transitory computer-readable media of claim 8, wherein the operations comprise performing a fault detection or diagnostic procedure for the building equipment based on the stability index.

12. The one or more non-transitory computer-readable media of claim 8, wherein modifying the operation of the building equipment comprises modifying a control strategy for the building equipment and operating the building equipment in accordance with the modified operation comprises operating the building equipment using the modified control strategy.

13. The one or more non-transitory computer-readable media of claim 8, wherein the building equipment comprises a chiller and the operations comprise calculating a chiller performance index.

14. The one or more non-transitory computer-readable media of claim 8, wherein comparing the stability index to the criterion comprises determining whether the stability index is greater than a predefined threshold.

15. A building management system, comprising:
building equipment operable to affect a condition of a building based on a setpoint for the condition;
a sensor configured to measure an actual value of the condition of the building;
a computing resource configured to:
calculate a stability index based on a timeseries of errors between the actual value and the setpoint;
compare the stability index to a criterion; and
in response to the stability index satisfying the criterion, modify an operation of the building equipment and operate the building equipment in accordance with the modified operation.

16. The building management system of claim 15, wherein the condition is humidity, pressure, or flowrate.

17. The building management system of claim 15, wherein the computing resource is configured to calculate the stability index by calculating a sum of a plurality of trapezoidal areas defined by the timeseries of errors.

18. The building management system of claim 15, wherein the computing resource is configured to perform a fault detection or diagnostic procedure for the building equipment based on the stability index.

19. The building management system of claim 15, wherein the computing resource is configured to modify the

operation of the building equipment by modifying a predictive control strategy for the building equipment.

20. The building management system of claim **15**, wherein the computing resource is configured to compare the stability index to the criterion by determining whether the stability index is less than a predefined threshold.

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